

Ubiquitous predictive processing in the spectral domain of sensory cortex

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The authors analyzed spectral properties of neural activity recorded using laminar probes while mice engaged in a global/local visual oddball paradigm. They found **solid** evidence for an increase in gamma (and theta in some cases) for unpredictable versus predictable stimuli, and a reduction in alpha/beta, which they consider evidence towards a "predictive routing" scheme. The study is overall **important** because it addresses the basis of predictive processing in the cortex, but some of the analytical choices could be better motivated, and overall, the manuscript can be improved by performing additional analyses.

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Abstract

The appearance at the anatomical level of a canonical laminar microcircuit suggests that each six-layer column of granular cortex may mediate a canonical computation. Hypotheses for such computations include predictive coding, predictive routing, efficient coding, and others. However, single-neuron recordings capture only the individual elements of the hypothesized laminar microcircuit, while local field potentials (LFPs) from a laminar probe offer insight into the broader population activity. Through the Allen Institute's OpenScope Brain Observatory, data in mice performing a visual oddball task during multi-area laminar recording was used to test predictive processing hypotheses in the spectral domain. Histological labeling of the cortical laminae enabled a fine-grained examination of their roles in the task, and frequency bands capturing both feedforward and feedback effects were analyzed. γ -band local-field potential (LFP) oscillations conveyed feedforward prediction errors in lower sensory areas of cortex; α/β -band oscillations weakened in unpredictable conditions compared to predictable ones; and θ -band oscillations additionally signalled slower, longer-scale temporal prediction errors. In combination with the previous findings, predictive routing explains these experiments where neither ubiquitous predictive coding nor feedforward adaptation can.

Significance

Cortical columns robustly signal perceptual features through the firing rates of spiking neurons. In accordance with this rate coding, predictive processing theories hypothesized that neuronal firing rates ubiquitously signal surprise. However, a recent large-scale study of spike rates did not support this conjecture. An alternate model, predictive routing, suggests that neuronal oscillations rather than spike rates could encode surprise. These neuronal oscillations, which can affect the timing but not rate of spiking, formed coherent γ rhythms which consistently signaled both simpler and more complex forms of surprise in mouse visual cortex. Together with the findings on spike-rates in the same experiment, our findings suggest that cortical circuits encode surprise in the rhythmic timing of spikes rather than in their rate.

Main Text

The brain is continually bombarded with noisy, ambiguous sensory signals from which it must construct actionable percepts via an internal model. Oddball paradigms featuring repeated, or otherwise patterned, stimuli allow experimentation on the neural basis of this internal model. Responses to repeated stimuli decline with added repetition(1–4), and many measurement modalities that directly (e.g., the local field potential (LFP)) or indirectly (e.g. blood-oxygen level dependent (BOLD) responses(5)) derive from collective circuit activity also appear to reflect the predictability of stimuli(6). These observations have motivated the theoretical paradigm of *predictive processing*(7–9), founded upon early theoretical studies of predictive coding in the retina(10) and primary visual cortex(11).

Predictive coding (PC) hypothesizes that sensory perception consists of combining top-down and bottom-up signals in a canonical cortical microcircuit(12). PC models the top-down signals as “predictions” that serve to both explain away anticipated stimuli and to disambiguate uncertain stimuli, while hypothesizing that aspects of the stimulus not thus explained ascend the cortical hierarchy as bottom-up “prediction errors”. Experimentally, previous studies have used various auditory and visual oddball paradigms that first set up a predictable pattern, typically by means of stimulus or pattern repetition, and then violate it(13). Neuronal responses to the violations are interpreted as prediction errors.

The question of which neuronal circuits implement predictive processing in the brain remains open. Previous work has focused on decoding oddballs from evoked spiking rates(6, 14–17). A recent large-scale neurophysiological survey(17) of visual cortex in both mice and monkeys reported on the same visual oddball paradigm used here, but confined its analyses to the rate-coded spiking domain. While this previous study did find that some prediction errors were detected in higher-order areas (LM, PM, and AM in mice) and fed back down, it did not test PC models’ hypotheses (12, 18–21) in the spectral domain.

A growing body of work (22–28) has linked predictive processing to the oscillatory dynamics of LFPs. Presently, a controversy exists regarding which bands of cortical oscillations correspond to which theoretical constructs in predictive processing, specifically regarding the role of γ -band oscillations. Some authors(24, 29) have characterized γ oscillations as synchronizing the superficial layers of neighboring cortical columns according to the spatial predictability of the stimuli driving them, while other authors(25, 27, 30) have associated θ and γ oscillations with prediction errors and α/β oscillations with predictions. The latter accounts enjoy compatibility with findings(22, 31–34) that α/β and γ oscillations respectively mediate

feedback and feedforward signaling. We describe this account of feedback and feedforward signaling, with α/β exerting inhibitory control over γ in terms of the proposed general cortical mechanism of “predictive routing”(27 [↗](#)). Still other authors(35 [↗](#)) provided evidence that in a roving oddball paradigm, the prediction error signal appears in the aperiodic component of LFP spectrograms. Under this perspective, spectral changes resulting in increases in γ and decreases in α/β (or vice-versa) are better captured by aperiodic shifts rather than changes to specific neuronal oscillations(36 [↗](#)).

The contradictions between the above accounts require careful experiments that disentangle the different possible neuronal mechanisms purported to generate prediction-error signals, as well as analysis that takes into account the distinctions argued in the literature. This paper tests core tenets of the (hierarchical) predictive routing model using Multi-Area, high-Density, Laminar-resolved Neurophysiological (MaDeLaNe) recordings of local field potentials in mice; these mice underwent a large number of pre-recording repetitions of the “xxxxy” pattern displayed in the Global/Local Oddball paradigm (13 [↗](#)) to ensure that they would form perceptual expectations about the sequence and attenuate any novelty responses to the sequence. We analyzed data from six visual cortical areas, from the primary visual cortex (V1) to AM(37 [↗](#)) at the top of the mouse visual hierarchy. After examining oscillatory activity in the spectral space, we also wanted to know whether the oscillatory bands analyzed routed information in the feedforward or feedback directions within the cortical column. Previous work in macaque areas V1(27 [↗](#)) has identified a laminar pattern of γ current sinks feeding from cortical layer 4 to more superficial and deep layers, laminar pattern of activation classically associated with feedforward processing(12 [↗](#), 38 [↗](#)). Previous work was also found that α sinks originate in superficial and deep layers and move towards layer 4, a pattern classically associated with feedback processing. We used these same distinctions to test for laminar feedforward/feedback flow in the γ , α/β and θ frequency ranges during oddball processing.

These techniques enabled us to test the following core hypotheses of predictive routing in both local and global oddballs, as well as random deviants. **H1**: Predictive routing suggests that feedforward γ increases in the processing of unpredicted stimuli, while feedback α/β serves to selectively suppress the processing of predicted stimuli. This results in a “push-pull” dynamic between α/β and γ . H1 further suggests a role for θ oscillations in prediction error processing as well. **H2**: Counter to H1, the “ γ as prediction” model suggests that γ synchronization reflects the confirmation of predictions by an unsurprising stimulus, as suggested by previous physiology work in the spatial(24 [↗](#)) domain and modeling work across spatial(29 [↗](#)) and temporal(26 [↗](#)) domains. **H3**: Predictability vs unpredictability tilts or shifts the aperiodic component of the spectral response, rather than affecting oscillations within specific frequency ranges.

We tested H1-H3 by submitting the data to analysis during three different oddball types, which we term “deviance detection”, “global oddball”, and “stimulus-specific adaptation”. Importantly, which stimulus-specific adaptation tests how the brain responds to a change of stimulus, deviance detection and global oddball contrasts both test for effects of prediction while controlling for the effects of short-term adaptation. Consistent with **H1**, contrasting these three different types of oddballs against their corresponding controls demonstrated a clear, significant increase in γ oscillations between 30-60 Hz, although the timing, laminar origin, and center frequency of these responses differed across oddball types. Consistent with **H1** but inconsistent with **H2**, global oddballs increased γ oscillations, showing that violation of predictions nonetheless drove an increase in γ power. Partially consistent with **H3**, each oddball induced a significant spectral shift in at least one area, but only in the first 150 milliseconds of spectral response.

Results

Animals were first habituated extensively to local oddball sequences in two cohorts over several days (**Figure 1a**): one to an “xxxy” sequence and the other to a “yyyy” sequence. The ‘a’ and ‘b’ stimuli within a sequence denote drifting gratings at 135 or 45 degrees from horizontal; in the rest of the paper, ‘x’ will refer to the angle of stimulus used in the first three stimuli of the habituation sequence and ‘y’ to the angle of stimulus used in the last stimulus of the habituation sequence. Lower-case letters will denote fully predictable sequence elements (probability $P = 100\%$), while capital letters will denote randomized or oddball sequence elements (with the respective probability of occurrence defined in the text where relevant). The paper will proceed to describe the habituation sequence as “xxxy”. The global oddball sequence “xxxX” sequence did not occur during training, nor during the initial 50 habituation trials (**Figure 1b**) of the recording session. Only the main block contained the “xxxX” sequence in 20% (randomized) of its trials (**Figure 1b**).

To investigate the neural signaling underlying prediction of predictable sequences and processing of unpredicted stimuli, we compared responses in the main block with those in two control blocks (**Figure 1d**): a random block (in which “XXXX” had probability $P = 6.25\%$, the same as “XXXY”) and a fully predictable block (in which sequences were predictable from their first stimulus, so that “xxxx” had $P = 50\%$ and “yyyy” had $P = 50\%$). To analyze oddball responses, we compared spectral responses to the fourth stimulus in the sequence (denoted P4) across blocks. First, to examine modulation of responses by probabilistic context, we contrasted P4 in the randomized control block (‘Y’ in “XXXY”) with P4 in the fully deterministic habituation block at the beginning of recording (‘y’ in “xxxy”); we refer to this contrast as “deviance detection”. Second, to examine the response to prediction error without a release from stimulus-specific adaptation, we contrasted P4 in the global oddball (the ‘X’ in “xxxX”) against P4 in the deterministic control block (the last ‘x’ in “xxxx”). We refer to this contrast as the “global oddball”. Third, to examine the spectral effects of release from stimulus-specific adaptation, we contrasted P4 in the local oddball (the ‘Y’ in “xxxY”) against P4 in the deterministic control block (the last ‘y’ in “yyyy”). We refer to this contrast as “stimulus specific adaptation”.

After the spectral decomposition of the LFPs was performed, all contrasts consisted of a channel-time-frequency (three-dimensional) nonparametric, cluster-based statistical test (39) with a family-wise error rate $\alpha = 0.01$; the entire paper uses this significance level and testing procedure for its statistical tests. The tests thus resulted in three-dimensional significance masks for the difference in trialwise means of spectrograms. All reported results passed significance at this threshold (see Methods below for further details). We also performed the aperiodic vs oscillatory split analysis specifically for the first 150 ms of the spectral response, a feedforward sweep before feedback signaling comes online during stimulus presentation (40, 41); **Supplementary Figure 3** shows the results of this analysis and will be referred to throughout the presentation of results.

Narrowband γ oscillations reflect deviance detection

The Global-Local Oddball paradigm (13) contains the standard oddball sequence in two different probabilistic contexts: the habituation block’s “xxxy” sequences ($P = 100\%$) and the random control block’s fully randomized “XXXY” sequences ($P = 15\%$), the latter following the abolition of the habituation and therefore abolition of the highly predictable nature of “xxxY”. Combining these for our deviance detection contrast enabled testing **H1**, **H2**, and **H3** via the effects of context upon responses to the standardized oddball sequence.

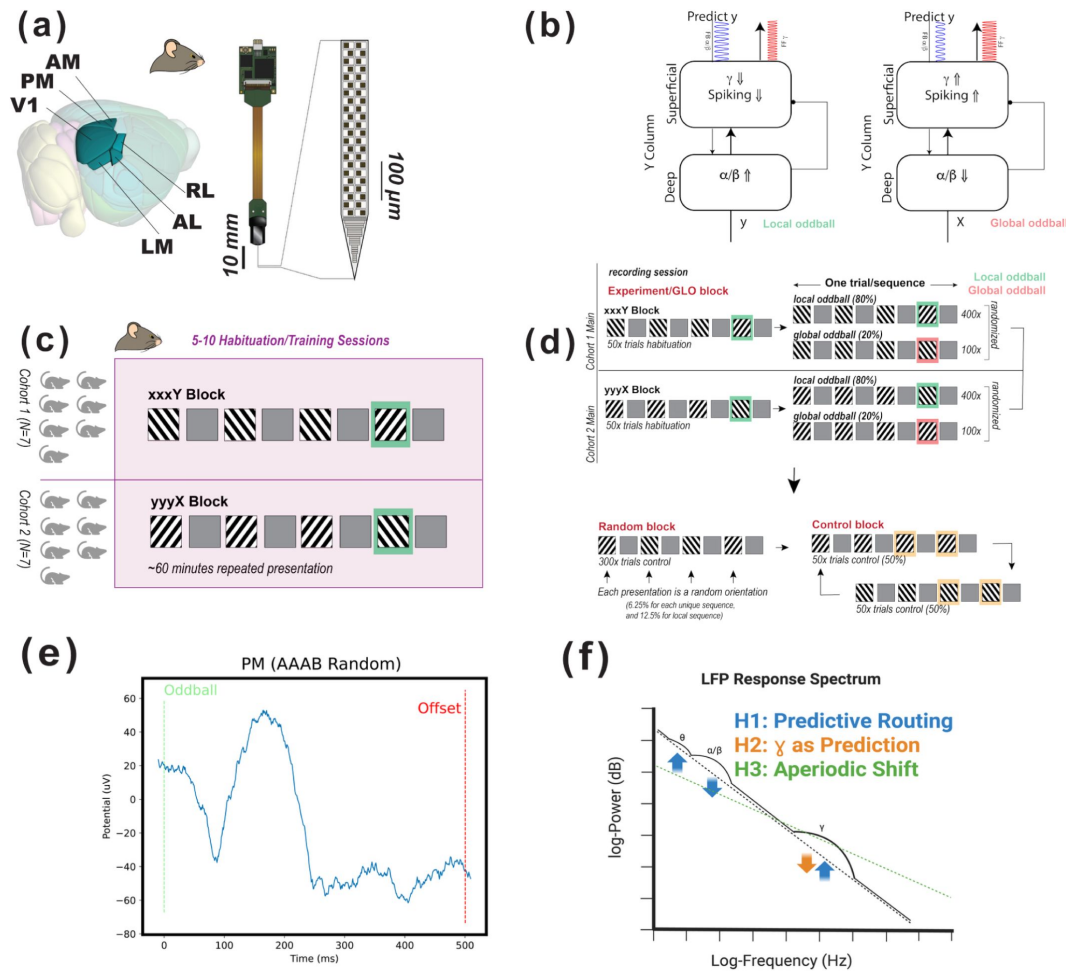


Figure 1.

Local/global oddball paradigm, experimental setup, and predictive routing.

(a) Neuropixels were introduced into 6 visual cortical regions in all mice (V1, LM, RL, AL, PM, AM). **(b)** How predictive routing models local and global oddballs: in the local oddball, a feedback α/β oscillation predicting stimulus 'x' will fail to inhibit cortical columns responsive to stimulus 'y', resulting in uninhibited γ oscillations and spiking upon the onset of an unexpected 'Y', while in the global oddball, a feedback α/β oscillation predicting stimulus 'y' after "xxx" will fail to inhibit cortical columns responsive to stimulus 'x', resulting in uninhibited γ oscillations and spiking upon the onset of the unexpected repetition 'X'. **(c)** N=14 animals were habituated to the local oddball sequences ("xxxY" in Cohort 1 of 7 animals, "yyyX" in Cohort 2 of 7 animals) before recording; stimuli were presented for 500 ms with a 500-531 ms inter-stimulus interval and 1 s inter-trial intervals. **(d)** Visual stimuli were presented in 4-stimulus sequences consisting of two (2) oriented, drifting bar gratings (symbolized as 'x' or 'y'), each displayed full-screen. After the main block of recording in which local and global oddballs were presented, the random control block consisted of randomized individual stimuli presented in sequences of four, while the repetition control block consisted of sequences of four repetitions of the same stimulus. **(e)** Local field potential in the middle channel of L4 over the course of the 'y' stimulus in the random deviant condition in the higher-order visual area PM; the plot shows oscillatory activity before, during, and after the stimulus-driven negative deflection in the local field potential. **(f)** Schematic of how surprise affects spectral responses according to hypotheses **H1** (predictive routing), **H2** (γ encoding prediction), and **H3** (aperiodic shifts). Dotted lines show the aperiodic component of the spectral response, while solid lines show the reconstructed spectrum that includes oscillatory peaks. **H1** (blue) predicts that surprise increases θ power, decreases α/β power, and increases γ power; **H2** (orange) predicts that surprise decreases γ power while confirmation increases it; and **H3** (green) predicts that surprise shifts the aperiodic component of the spectrum. Created in BioRender. Sennesh, E. (2025) <https://BioRender.com/ogllv24>.

H1 hypothesizes that unpredicted but attended stimuli evoke feedforward γ oscillations, while suppressing feedback α/β oscillations, and that probabilistic uncertainty, not just stimulus identity, modulates the power of these oscillatory responses, while **H2** hypothesizes that γ oscillations reflect confirmation of predictions (and thus, certainty rather than uncertainty). **H3** hypothesizes that predictability modulation should affect the aperiodic component of the spectral response, rather than the oscillatory component. Our experimental design enables testing these hypotheses with the three contrasts defined above. We note that a possibility is that γ , α/β , and aperiodic components may behave differently for each contrast, and alternatively, these signals may also behave consistently across the contrasts if they are canonically related to stimulus (un)predictability.

We begin with the deviance detection contrast in area V1. All analyses have been performed after removing the FOOOF-computed aperiodic component as a baseline, and are therefore more interpretable as genuine oscillatory activity. **Figure 2** shows the oscillatory spectral responses to the fully randomized final ‘Y’ presentation of the sequence “XXXY” ($P = 15\%$) in contrast to those to the fully deterministic ‘y’ presentation of the sequence “xxxy” ($P = 100\%$) during habituation. **Figures 2(a)** and **2(b)** show the spectrograms relative to the FOOOF-computed aperiodic component, while **Figure 2(c)** shows the spectrogram of the deviance detection contrast (see Methods for our channel pooling strategy of across trials and animals). We observed a characteristic decrease in α/β power (1-2 dB) just after stimulus onset, followed by γ oscillations (1-2 dB) that began at the 40-65 Hz frequency band and descended down towards the 35-55 Hz band over time. Since the cluster masks produced by our statistical tests are three-dimensional, **Figures 2(d-f)** display the laminar activation patterns corresponding to the full γ (30-90 Hz), α/β (10-30 Hz), and θ (2-10 Hz) bands. **Figure 2(d)** shows that γ oscillations are strongest in superficial layers 1-4 (around 2 dB increase in oddball-related power); after approximately 100 ms a weak (<1 dB) but significant loss of γ power occurs in laminar L5/6 while a gain in synchronization (1-2 dB) comes to dominate laminar L2/3. **Figure 2(e)** shows that increases in α/β power (1-1.5 dB in power) occurs throughout stimulus presentation in L1 of V1, though it is interrupted by a powerful (2 dB) desynchronization beginning in laminar L2/3 and spreading down into L4 and L5/6. **Figure 2(f)** shows that θ synchronization occurs only weakly (< 1 dB) in L1 of V1 during deviance detection. To determine the laminar origin of current sinks and sources of these LFP oscillations, and thus whether they index feedforward or feedback captivity, we analyzed the current source density relative to the troughs of these oscillations (see Methods for details). **Figures 2(g-i)** show the trough-locked current-source density profiles of the γ , α/β , and θ oscillations. We observed a feedforward pattern of activation in γ oscillations, whereby sinks originated in the middle layers and spread out to superficial L2/3, followed by later sinks in L5/6. There was a characteristic laminar feedback pattern in α/β oscillations, which originated in the most superficial contacts and spread to the middle layers. θ oscillations displayed a mixed pattern consistent with aspects of both a feedback and feedforward pattern.

To ensure that apparent deviance detection results in fact reflect neuronal oscillations, we also separated the spectrum within each condition for each animal into its aperiodic and oscillatory components via FOOOF; **Supplemental Figures 1** and **2** show these results. Deviance detection displayed no statistically significant difference between the aperiodic components of the spectral response to its oddball and its control stimulus in any of the recorded cortical areas. After removal of the aperiodic component, however, only V1 ($p < 0.05$) and LM ($p < 0.01$) displayed significant increases in both the power and peak frequency of γ oscillations, derived from a permutation-based t -test (with $n = 1000$ permutations) for difference of means between conditions on per-subject, channel- and trial-averaged frequency spectra. **Supplementary Figure 3** shows that deviance detection induced a significant difference in the aperiodic spectral response of the feedforward sweep of 150 ms only in LM.

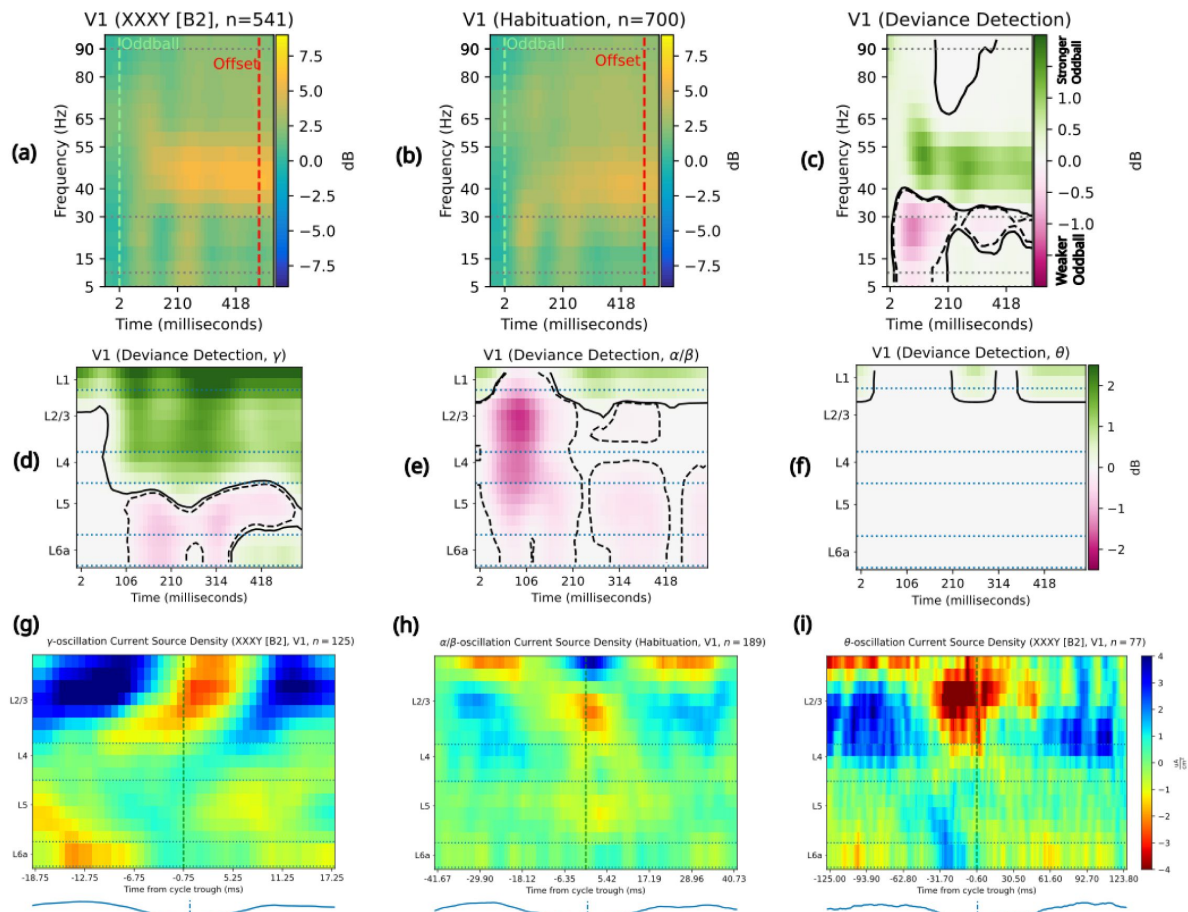


Figure 2.

Probabilistically deviant stimuli induce a “push-pull” between α/β - and γ oscillations.

The top row shows the two conditions contrasted to examine deviance detection; the middle row shows the laminar signatures of the contrast results broken out by frequency band; and the bottom row shows the laminar CSD responses characteristic of each frequency band in these conditions. Solid contour lines enclose the area of a synchronization/power increase, while dotted contour lines enclose the area of a desynchronization/power decrease. **(a)** Spectral responses to a fully randomized “XXXX” ($n=541$ trials) conform to stereotyped “oddball” responses, with stimulus onset leading to an increase (2.5-5 dB) and then decrease (0-2.5 dB) in α/β power over baseline, then to a long-lasting increase (2.5-5 dB) in γ (40-55 Hz) synchronization. **(b)** Spectral responses to a fully predictable, habituated “xxxx” ($n=700$ trials) show a fast increase (>5 dB) over baseline and then return to baseline of α/β power, followed by a long-lasting increase (2.5-5 dB) in γ (40 rising to 40-55 Hz over time) synchronization. **(c)** Detection of probabilistic deviants, via the contrast between “XXXX” and “xxxx”, induces a 1 dB decrease in α/β power and a 1-2 dB γ power increase. **(d)** Deviance-induced γ (30-90 Hz) synchronization occurs earliest and most strongly at the boundary between L1 and L2/3, remaining strong in L1 while spreading down into L2/3; a 0-1 dB loss of γ power occurs in L5/6 100ms into deviance detection. **(e)** Deviance-induced α/β (10-30 Hz) activity starts with a fast-onset increase in synchronization in L1, followed by α/β desynchronization (1.5-2 dB) in L3 and down into L5, followed by a slow decrease in α/β power (0.5 dB) in L5/6 and increase in α/β power (1-1.5 dB) in L1. **(f)** Deviance detection induces a weak (0.5-1 dB), late increase in θ (2-10 Hz) power. **(g)** γ oscillations in P4 of the fully randomized “XXXX” sequence display a canonical feedforward pattern of laminar activation, beginning in L4, then flowing out to L3 and later into L5/6, over $n=125$ complete oscillatory cycles re-epoching from 541 trials. Oscillatory LFP trace shown below for illustrative purposes only. **(h)** α/β oscillations in P4 of the fully habituated “xxxx” sequence display a weak but canonical feedback pattern of laminar activation, beginning in L1 and flowing down to L2/3, then down through L4 to meet a rising flow of activation in L5. CSD was calculated over the average of $n=189$ complete oscillatory windows over 700 trials. Oscillatory LFP trace shown below for illustrative purposes only. **(i)** θ oscillations in P4 of the fully randomized “XXXX” sequence display a canonically feedback pattern of laminar activation, beginning in L1 and L6 before converging up through L4 to L2/3, followed by a feedforward pattern to L1 and L5/6 from L2/3. CSD was calculated over the average of $n=77$ complete oscillatory cycles out of 541 trials. Oscillatory LFP trace shown below for illustrative purposes only.

These findings show consistency with hypotheses **H1** and inconsistency with **H2**. Consistent with predictive routing (**H1**), γ power increased, α/β power decreased, and θ power increased to an unpredictable stimulus vs a predictable one. Inconsistent with **H2**, the random deviant “XXX γ ” exhibited higher, rather than lower, γ power than the fully predictable “xxx γ ” in habituation. **H3** concerns the aperiodic component of the spectral response, so combined with the results in Supplemental Figures 1, 2, and 5 these results are mostly inconsistent with **H3**, excepting the difference in LM in the aperiodic component.

γ oscillations responded to global oddballs

The Global-Local Oddball paradigm (13) contains two conditions in which stimulus repetition occurs without a release from neuronal adaptation: during the main block, the global oddball “xxx γ ” occurred with $P = 20\%$, while during the repetition control block, “xxxx” occurred during half the trials and its final stimulus ‘ γ ’ occurred with $P = 100\%$ given the first stimulus. This global oddball contrast enabled testing **H1**, **H2**, and **H3** without the effects of any release from stimulus-specific adaptation.

Figure 3 shows the oscillatory spectral responses, measured in decibels of difference across conditions, to the global oddball “xxx γ ”, in contrast to those to the deterministic control sequence “xxxx”. **Figure 3(a)** shows the response to the final ‘ γ ’, while **Figure 3(b)** shows the response to a predictable ‘ γ ’ in the control condition. Both responses consist of an increase in α/β power over baseline at the onset of the stimulus, alongside an increase in γ (30-90 Hz) power in 40-55 Hz and 80-90 Hz bands that lasts throughout stimulus presentation. However, in the global oddball condition the α/β band quickly returns to baseline, while the γ oscillations increase more in power (2 dB vs <2 dB) compared to the control block. **Figure 3(c)** shows the statistically significant contrast between the global oddball ‘ γ ’ and stimulus control ‘ γ ’, derived from a channel-time-frequency nonparametric statistical test with family-wise error rate $\alpha = 0.01$ in which spectrogram data was averaged over channels; the contrast response consists of a fast increase (≥ 0.3 dB) in high- γ (55-90 Hz) synchronization, which trails off into sustained increases (≤ 0.2 dB) in 40-55 Hz γ power, alongside an immediate rise (0.2-0.3 dB) in θ power and a fall (<0.2 dB) in α/β power over time. We interpret the low-frequency synchronization shown in **Figure 3(c)** as θ synchronization in light of a further spectral analysis with higher frequency resolution, shown in Supplemental Figure 2, finding oscillatory peaks at 5-9 Hz, within the <10 Hz θ band. **Figures 3(d)**, 3(e), and 3(f) each display one frequency band’s laminar profile over time. **Figure 3(d)** shows that an immediate increase in γ (0.4-0.6 dB) power occurred in the supragranular laminar layers and grew to ≥ 0.6 dB over the course of global oddball presentation. **Figure 3(e)** shows that an immediate α/β loss of power (0.75-1 dB) took place in upper supragranular layers, while a weak increase in α/β power spread from infragranular layers upwards after approximately 120 ms. **Figure 3(f)** shows that within less than 100 ms after oddball onset, an increase in θ power (<0.5 dB) begins in infragranular layers before intensifying to 1.0-1.5 dB as it spreads throughout V1. **Figures 3(g-i)** show the current source density profiles of the γ , α/β , and θ oscillations, time-locked to the oscillation troughs; these demonstrate a pattern of feedforward activation in γ oscillations, a feedback pattern in α/β oscillations, and a pattern of feedback (L5/6 to L4) followed by feedforward (L4 to L2/3 and L5/6) activation in θ oscillations.

Ensuring that apparent global oddball results in fact reflect neuronal oscillations, the aperiodic component of the per-animal, per-condition spectra showed no significant differences between the last response in “xxx γ ” and “xxxx” sequences (**Supplementary Figure 1**), although significant spectral tilt does appear fast feedforward sweep in the first 150 ms (**Supplementary Figure 3**) in V1, AL, and PM. In fact, the primary significant differences driven by the global oddball were in the more predictable condition “xxxx” showing greater α/β than the global oddball “xxx γ ” (**Supplementary Figure 2**) in V1, LM, RL, AL, and PM. We therefore interpret our global oddball results in terms of both spectral tilt and oscillation dynamics.

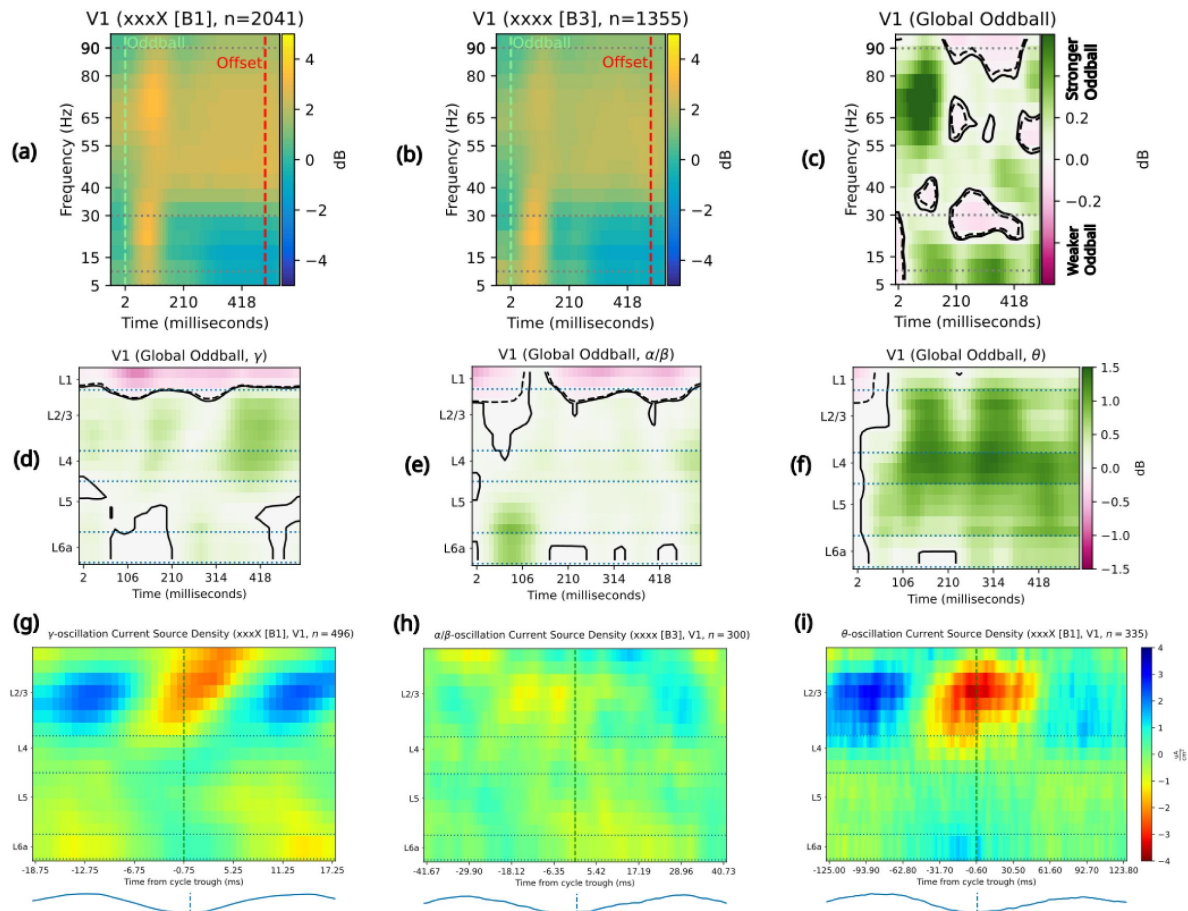


Figure 3.

Unpredictable repetitions induce γ and θ oscillations.

The top row shows the two conditions contrasted to examine global oddball detection; the middle row shows the laminar signatures of the contrast results broken out by frequency band; and the bottom row shows the laminar CSD responses characteristic of each frequency band in these conditions. Solid contour lines enclose the area of a synchronization/power increase, while dotted contour lines enclose the area of a desynchronization/power decrease. **(a)** Spectral responses to P4 in an unpredicted (20% probability, $n=2041$ trials) repetition “xxxX” show a fast increase (2-4 dB) in α/β (2-4 dB) and increase (<2 dB) in high- γ power, followed by a return to baseline in the α/β band, a <2 dB increase in θ synchronization, and long-lasting <2 dB increases in low- γ (40-55 Hz) and high- γ (80-90 Hz) synchronization. **(b)** Spectral responses to P4 in a fully predictable ($n=1355$ trials) repetition sequence “xxxx” show a fast increase (4-6 dB) in α/β power and smaller (<2 dB) γ power increase over baseline; both bands of oscillations continued throughout the first 200ms. **(c)** Global oddball detection, via the $\alpha = 0.01$ statistical contrast between “xxxX” and “xxxx”, induces a 0.2 dB decrease in α/β power after 200ms, alongside 0.4-0.5 dB increases in θ and γ power. **(d)** γ power increases by 0.2-0.4 dB in L5/6, L4, and L2/3 immediately upon onset of the global oddball, persisting and even strengthening to 0.6 dB by the end of stimulus presentation, while γ power decreased by 0.4-0.6 dB in L1. **(e)** α/β power increase begins (0.5-1 dB) in L5/6 <100ms after global oddball onset, while α/β power decreases (0.5-0.75 dB) in L1. The increase in infragranular layers spreads out to L4 and then to L2/3 while weakening to 0.25-0.5 dB. **(f)** θ oscillations begin in L5/6 with a 0.5dB increase just after global oddball onset before spreading and strengthening (1-1.5 dB) throughout the laminar column. **(g)** γ oscillations induced by global oddballs show a canonically feedforward laminar activation pattern, starting in L4 at phase 0 and spreading out towards L2/3 and L5/6 as the phase cycle continues. CSD was calculated over the average of $n=496$ complete oscillatory windows over 2041 trials. **(h)** α/β show a weakly feedback-driven laminar activation pattern, starting in L1 before continuing down to L2/3 and L4, with later contributions coming from L5/6. CSD was calculated over the average of $n=300$ complete oscillatory windows over 1355 trials. **(i)** θ oscillations show a feedforward laminar activation pattern under global oddballs, with activation beginning in L5 and L4 before spreading to L2/3 as it strengthens with the phase cycle. CSD was calculated over the average of $n=335$ complete oscillatory windows over 2041 trials.

These findings are consistent with **H1**, inconsistent with **H2**, and partially consistent with **H3**. The pattern of combined γ and θ synchronization with α/β desynchronization, driven by an unpredictable stimulus, fits **H1**. Both the global oddball and control conditions feature identical stimulus sequences, but the γ power increase for the surprising ($P = 20\%$) global oddball rather than predictable ($P = 100\%$) control is inconsistent with **H2**. **H3** hypothesized that unpredictability should affect the aperiodic component of the spectrum rather than the oscillatory one; the findings in [Figure 3](#) combine with those in [Supplementary Figure 3](#) to show consistency with **H3** for the fast feedforward sweep of 0-150 ms post-onset, while combining with those in [Supplementary Figure 1](#) to falsify **H3** for the later 150-500 ms period.

Stimulus change induces γ vs α/β push-pull

H1 hypothesizes that classic, “local” oddballs should induce a push-pull dynamic between feedforward γ and θ synchronization and α/β desynchronization. **H2** suggests that γ synchronization should reflect the confirmation of predictions by the stimulus. **H3** suggests that predictability or unpredictability should shift the aperiodic component of the spectral response rather than the oscillatory one. The Global-Local Oddball paradigm ([13](#)) enables testing for “local” oddballs (in which repetition creates an expectation which is violated via a change of stimulus content) for the effects of a release from stimulus-specific neuronal adaptation. This test consists of contrasting the oscillatory responses to a “local” oddball sequence “xxxY”, in which a release from adaptation would occur with stimulus change, with those to fully deterministic sequences having the stimulus content “yyyy”, in which neuronal adaptation would continue throughout all four presentations (see [Figure 1\(d\)](#) for a full description of sequence types).

[Figure 4\(a-c\)](#) shows the oscillatory spectral responses, measured in decibels above the aperiodic baseline, to the mostly predictable ($P = 80\%$) final ‘Y’ presentation of the sequence “xxxY” ([Figure 4\(a\)](#)) in contrast to those of the fully deterministic ($P = 100\%$) ‘y’ presentation of the sequence “yyyy” ([Figure 4\(b\)](#)). The contrast in [Figure 4\(c\)](#) shows non-frequency-specific desynchronization (0.5-1.5 dB) at the onset of the local oddball, followed by enhanced γ power (1-1.5 dB) throughout the stimulus presentation. [Figures 4\(d-f\)](#) each show the laminar profile of significant contrasts in oscillatory power within the γ , α/β , and θ frequency bands. This confirms that the desynchronization is present in all frequency ranges and layers.

Additionally, [Figure 4\(d\)](#) shows a γ power increase (1-2 dB) that starts in L5/6 and spreads up through L4 into L2/3 and L1. [Figure 4\(e\)](#) shows that an increase in α/β power (1 dB at its strongest, at approximately 150 ms) starting in L5 and spreading out to the rest of the cortical column. [Figure 4\(f\)](#) shows a θ synchronization (1-1.5 dB) beginning in L5/6 and spreading up into L4 and through to L2/3 by the time of local-oddball offset. [Figures 4\(g-i\)](#) show the laminar current-source density profiles of the γ , α/β , and θ oscillations, time-locked to the oscillation troughs; these demonstrate a pattern of feedforward activation in γ oscillations ([Figure 4\(g\)](#)), a mixed feedforward and feedback pattern in α/β oscillations ([Figure 4\(h\)](#)), and a pattern of canonical feedforward (L4 to L2/3 and L5/6) activation in θ oscillations ([Figure 4\(i\)](#)).

These findings appear most consistent with **H1**, partly inconsistent with **H2**, and partially consistent with **H3** (in light of the findings in [Supplemental Figure 1](#) and [Supplementary Figure 3](#)). The push-pull effects between α/β and γ oscillations conform to **H1**, but also motivated it. Partly inconsistent with **H2**, the change of stimulus involved in a local oddball increased γ power after the first 150 ms, rather than decreasing it; however, a γ increase in response to the local oddball ($P = 80\%$) admits an explanation in terms of prediction confirmation via **H2**. Spectra for the full stimulus presentation displayed no significant effects of release from adaptation, inconsistent with **H3**, but spectra for the fast feedforward sweep in the first 150 ms, typically accompanied by high spiking, showed significance tilt in V1, AL, and PM, consistently with **H3** for that early sweep.

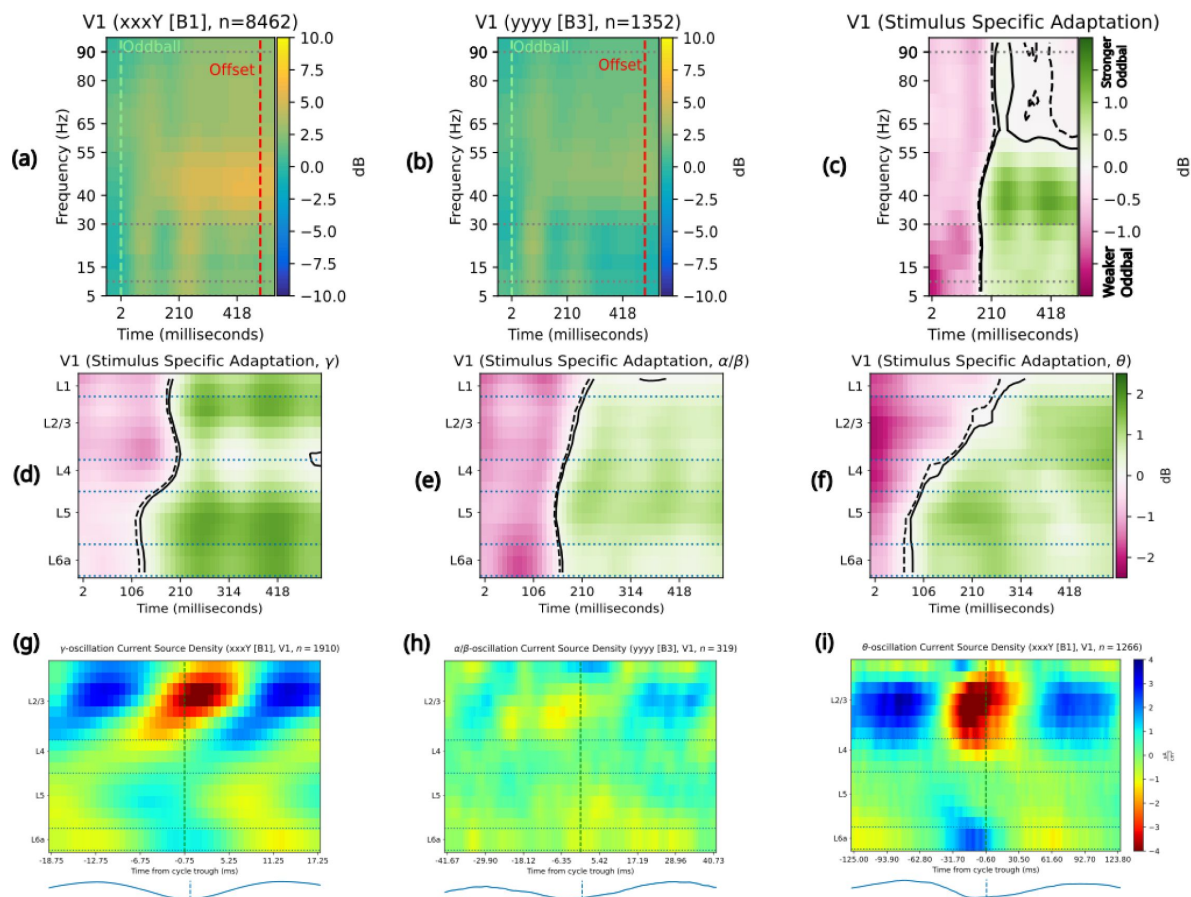


Figure 4.

Release from stimulus-specific adaptation induces a “push-pull” between α/β - and γ oscillations.

The top row shows the two conditions contrasted to examine release from stimulus-specific adaptation; the middle row shows the laminar signatures of the contrast results broken out by frequency band; and the bottom row shows the laminar CSD responses characteristic of each frequency band in these conditions. Solid contour lines enclose the area of a synchronization/power increase, while dotted contour lines enclose the area of a desynchronization/power decrease. **(a)** Spectral responses to P4 in a relatively predictable (80% probability) “xxxY” sequence conform to stereotyped “oddball” responses, with stimulus onset leading to an increase (2.5-5 dB) in α/β power (followed by a return to baseline and then another, more sustained increase), accompanied by a fast-onset and then long-lasting increase (2.5-5 dB) in γ (40-55 Hz) and high- γ (80-90 Hz) synchronization. **(b)** Spectral responses to P4 in a fully predictable “yyyy” sequence show a fast increase (2-4 dB) in α/β power after stimulus onset, followed by a smaller (<2 dB) increases in γ (40-55 Hz) and high- γ (80-90 Hz) synchronization. **(c)** Mostly-predictable release from adaptation, via the $\alpha = 0.01$ statistical contrast between “xxxY” and “yyyy” sequences, induces a 1-2 dB decrease in α/β power at stimulus onset, followed by a 1-2 dB γ power increase and a roughly 1 dB increase in θ and α/β power. **(d)** Release from stimulus-specific adaptation induces a fast, broad loss (<1 dB) of γ power, followed after 150ms by a 1 dB γ power increase from baseline that begins in L5/6 and spreads through L4 to L2/3. **(e)** Release from stimulus-specific adaptation induces a fast, broad loss (1-2 dB) of α/β power, followed by a 1-2 dB increase from baseline in α/β power that starts in L5/6 and spreads into L4 and L2/3 while never arriving to L1. **(f)** Release from stimulus-specific adaptation induces a fast, broad loss (1.5-3 dB) of θ synchronization that lasts longer in L2/3 than elsewhere, while after about 100ms a 1 dB increase in θ synchronization begins in L5/6 and spreads into L4 and L2/3 without ever arriving to L1. **(g)** γ oscillations in P4 of a mostly predictable “xxxY” sequence shows a canonical feedforward pattern of laminar activation, beginning in L4 and L5 at the end of one phase cycle and spreading out into L2/3 and L6 through the next cycle until the γ trough. CSD was calculated over the average of $n=1910$ complete oscillatory windows over 8462 trials. **(h)** α/β oscillations in P4 of the fully predictable “yyyy” sequence show a feedback pattern of laminar activation, with current sinks spreading from L5/6 and L1 into L5, L4 and L2/3 before the end of the oscillation carries them into L1 and L6. CSD was calculated over the average of $n=319$ complete oscillatory windows over 1352 trials. **(i)** θ oscillations in P4 of the mostly predictable “xxxY” sequence show a mixed feedback and then feedforward pattern of laminar activity, with current sinks starting in L1 and L6 before activity in L2/3 at the θ trough spreads downward through L4 and L5 into L6 towards the cycle’s end. CSD was calculated over the average of $n=1266$ complete oscillatory windows over 8462 trials.

Spectral power effects come from oscillations and aperiodic tilt

Spectra of neuronal local field potentials consist not only of oscillations but also of an aperiodic component whose strength tends to follow a $1/f$ law; without separating out this component from oscillations, false conclusions might be drawn (42). **Supplementary Figures 1** and 2 show the results of applying FOOOF analysis to parameterize LFP spectra across the period of oddball processing (from 50 ms before oddball onset to 50 ms after oddball offset) within conditions and subjects into aperiodic and oscillatory components. Supplemental Figure 1 shows the aperiodic components calculated for oddball spectra calculated for each animal and condition, with the line showing the mean aperiodic component across animals and the error bands showing two standard errors of the mean across animals; no significant cross-condition differences appear in the aperiodic component. **Supplementary Figure 2** shows that after removing the aperiodic component, oscillatory peaks in the power spectrum at θ and γ ranges exist in all areas and conditions. Interestingly, we observed a spectral shift in the γ frequency band most notably in the deviance detection contrast. We found that γ shifted on average 3 Hz frequency during the XXXY condition (mean γ peak in V1: 38 Hz with 99% confidence interval 35.2–40.5 Hz) compared to the same stimulus sequence during habituation trials (mean γ peak in V1: 35 Hz, 99% confidence interval 32.6–36.8). **Supplementary Figure 3** shows that in the fast feedforward sweep of the first 150 ms after the stimulus presentation, V1, LM, AL, PM, and AM all exhibited a spectral tilt due to oddballs in at least one contrast, though no area showed a spectral shift in every contrast. Particularly in the case of the release from stimulus-specific adaptation, the spectral tilt in V1, PM, and AM fits with **H3**, despite the cross-condition contrasts in **Figures 2–4** concerning the oscillatory activity after FOOOF removes the aperiodic spectrum.

Finally, we tested to what extent these effects in V1 could generalize to other visual cortical areas in mice. We analyzed γ -band oscillatory power in L2/3 across areas, and this demonstrated that all three types of oddballs show significant increases in γ power for oddball over control across multiple areas of the cortical hierarchy (with area RL in global oddballs being the one exception, **Figure 5**). Deviance detection and global oddball modulations were restricted to the early part of the hierarchy (areas V1, LM, RL, and AL for deviance detection and areas V1, LM, and RL for global oddball). Stimulus specific adaptation responses could be seen throughout the entire hierarchy of areas. These results are compatible with **H1**, because they show that γ acts as a signature of prediction error for multiple contrast types and in multiple areas. **H2** could be supported only by a single area and contrast, as area RL showed significant γ desynchronization during oddballs.

Discussion

This paper aimed to test the hypotheses **H1–H3** about the oscillatory dynamics of visual cortical areas in the Global/Local Oddball paradigm (13) in mice. In every contrast of experimental conditions, unpredicted stimuli evoked increased γ power in superficial (L1–3) layers, with a feedforward current-source density pattern (✓ **H1**). All three oddball types also evoked α/β desynchronization (✓ **H1**), although this occurred later and with a different laminar pattern in global oddballs than in deviance detection and release from adaptation. The deviance detection (randomization of stimulus change) and global oddball (unpredicted repetition) contrasts showed increased γ power with the violation, rather than confirmation, of predictions (✗ **H2**). The release from adaptation contrast showed fast (0–150 ms) decreases in γ power upon onset of a prediction-confirming stimulus (✗ **H2**), but this was followed by increased γ power after 150 ms (✓ **H2**). While response spectra across the full half-second stimulus presentation did not show an overall spectral tilt, those for the first 150 ms, during which feedforward spiking takes place, did show a spectral tilt in several areas and contrasts (✓ **H3**); differences in the late (150–500 ms) responses were confined to the per-trial oscillatory spectra and their dynamics.

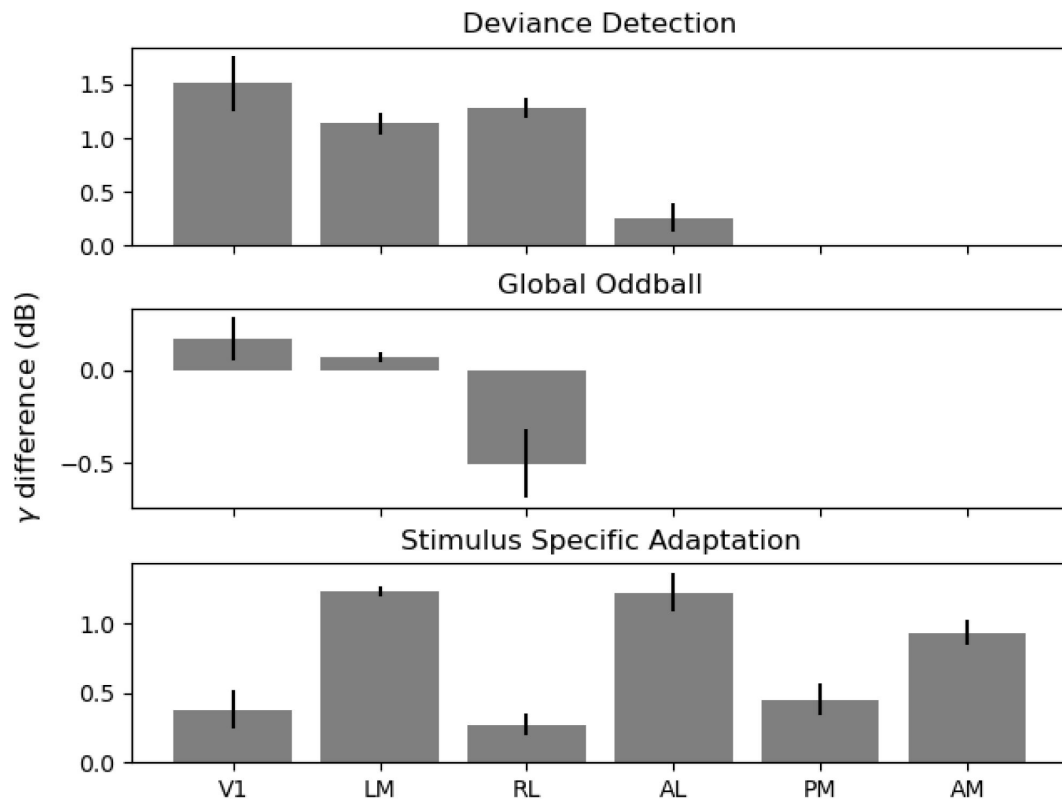


Figure 5

Early areas show greater, more significant differences in oscillatory γ power across deviance detection and global oddballs.

The bar-plot above shows the time-average of γ power increases/decreases in the statistically significant clusters ($\alpha = 0.01$) shown in **Figures 2(c)**, **3(c)**, and **4(c)** above bars show the mean (over channels) increase/decrease for each contrast while the error marks show two standard errors of the mean (over channels). Deviance detection induced a significant increase in oscillatory γ power of roughly 1.5 dB in V1, 1.1 dB in LM, 1.3 dB in RL, 0.3 dB in AL, and insignificant differences in PM and AM. Global oddballs induced smaller, but statistically significant (for channelwise statistics) increases in γ -band power of 0.2 dB in V1, 0.1 dB in LM, and -0.5 dB in RL, with no significant contrasts in AL, PM, and AM. Release from stimulus-specific adaptation generated significant increases in γ -band power across area, averaging to 0.4 dB in V1, 1.2 dB in LM, 0.3 dB in RL, 1.2 dB in AL, 0.4 dB in PM, and 0.9 dB in AM.

Our results suggest that the feedback role of α/β oscillations (see **Figures 2(h)**, 3(h), and 4(h)) likely consistently involves layers L1/6. Since L1/6 are already responsible for delivering feedback from a higher cortical area in canonical microcircuit models(43), this revision mainly considers the laminar profile of α/β oscillations rather than the participation of α/β oscillations in feedback processing. The predictive routing model(44) had previously considered α/β oscillations to come from L5/6 rather than L1/6, so ultimately our results suggest revising this to include a participation of both superficial and deep layers in α/β generation(45, 46). This participation of both the supragranular (L1) and infragranular (L6) targets of corticocortical feedback in α/β generation may be one reason why mice lack the “spectrolaminar motif” shown in primates(47).

The predictive routing model(6, 44) also observed θ oscillations in the course of afferent prediction-error processing, a feature found in all our oddball results but especially in the global oddball. This paper’s results suggest that γ synchronization routes fast transmission of feedforward stimulus information in feature-specific cortical columns(48) not inhibited by feedback α/β synchronization; the signature of global oddballs could indicate that α/β -band feedback silenced ‘Y’-responsive cortical columns, leaving ‘X’ columns to send fast γ -synchronized prediction-error signals, while also integrating longer-term sequence information into the slower θ -synchronized prediction error signal. This agrees with previous results (49) suggesting that θ oscillations may play a role in supporting sequence learning and recognition. It also fits with proposals whereby θ oscillations synchronize distant areas, enabling them to exchange relevant signals during cognitive processing(50, 51). Similarly, deviance detection and release from adaptation effects showed that the peak γ frequency shifted up for less predictable sequences compared to more predictable sequences (**Supplementary Figure 2**), another potential update to the predictive routing model of predictive processing.

Where this paper found evidence for predictive routing across multiple oddball types, recent work(17) analyzing the same dataset in the spike-rate domain did not find evidence for predictive coding in V1 across these same oddball types. How can population-level neuronal oscillations encode prediction errors when spike rates do not? Classical experiments(52–55) have found that information can be more strongly encoded in the timing of individual spikes as compared to the spike rate. Computational models(56–59) have captured such effects, further fortifying the neurobiological plausibility of temporal coding. Newer experiments have begun to suggest that ensemble temporal codes may reveal task-modulation effects beyond those reflected in spike rates and remain robust against representational drift(60, 61). Both the isocortex(58, 62–64) (hippocampal-entorhinal complex) and neocortex(65–68) show evidence of coding information in spike times via their relationship to the phase of a circuit-level oscillation. These results together suggest a hypothetical complimentary code for representing the brain’s internal mode, with spike rates encoding stimulus representation and oscillatory synchronization encoding relative uncertainty or confidence in these representations.

Population-level modeling studies have proposed that γ synchronization relates to the gain of superficial pyramidal neurons(20, 69, 70); through the lens of predictive processing, γ oscillations may regulate feedforward prediction-error signaling(12, 20, 21). This paper’s results show general consistency with these proposals. γ frequency shifts have also been proposed to enhance cortical gain(71). Frequency shifts have been observed in the context of attention and contrast(19, 70, 72, 73). Faster γ could function to have stronger downstream impact on receiving neurons(71). In our data, γ shifts to a significantly faster frequency during deviance detection (contrast DD) in V1 and LM (**Supplementary Figure 2**), supporting this proposal. We speculate that the deviant sequences attracted more bottom-up attention compared to the fully habituated and thereby predictable sequences. Release from stimulus-specific adaptation (contrast SSA, **Supplementary Figure 2**) also showed that in the sequence with higher overall uncertainty (the “yyyy” sequence in the control block with probability $P = 50\%$), γ shifted to a significantly higher peak frequency compared to the sequence with lower uncertainty (the “xxxY” in the main block with probability $P = 80\%$) in LM, RL, AL, and AM. We speculate that

again the less probable sequences attracted more bottom-up attention compared to the fully habituated, predictable sequences early in recording, and that understanding a stimulus change with probability $P = 80\%$ as a non-oddball may help to explain these results. Taken together, our results on enhanced γ power and faster γ frequencies are consistent with these prior theoretical proposals, suggesting that γ may play a role in the gain of sensory responses, although **Supplementary Figure 4** shows that despite SSA and GO contrasts modulating pupil size in Westerberg et al⁽⁷⁴⁾, DD did not seem to modulate pupil size and may not have modulated synaptic gain.

Complimentary coding of stimulus content in spike rates and surprisal in spike-timing or spike-field phase coherence (as found with γ oscillations and prediction errors in prefrontal and anterior cingulate cortex of macaques⁽⁷⁵⁾) would reconcile this paper's results, and the predictive routing model, with the evidence against spike-rate coding of prediction errors in sensory cortex⁽¹⁷⁾, although we leave measuring the relationship between the oscillations documented here and spike timing to future work. Since the local field potentials whose oscillations we analyzed here reflect neurons' membrane potentials^(76–78), this paper's results remain compatible with theoretical models of predictive processing^(79–82) in which membrane potentials, rather than spiking activity, encode the prediction-error quantity used to infer a rate-coded, evoked spike-train representation of the stimulus. This representation could consist of either samples or summary statistics of the posterior distribution while remaining compatible with existing experimental results, while spontaneous firing rates would hypothetically reflect similar representations of the prior distribution^(26, 83, 84) over previously-experienced stimuli.

This paper's results support, but also entail updating, the predictive routing model, in which surprise modulates α/β oscillatory power down and γ oscillatory power up. These results have the potential to rescue many aspects of predictive processing hypotheses drawn from recording or neuroimaging modalities driven in significant part by local field potential oscillations, complimenting the results of Westerberg et al⁽¹⁷⁾ showing spike-rate coding of surprise in some areas in macaque and mice with an oscillatory routing mechanism^(85, 86) for surprising stimuli in lower-order cortex.

Materials and Methods

Animals

All experiments were performed in 16 SSTAi32 and PVAi32 mice (*Mus musculus*) of both sexes ($n_{\text{male}}=6$, $n_{\text{female}}=10$), aged 95–128 days. All experiments on animals were conducted with the approval of the Allen Institute's Institutional Animal Care and Use Committee.

Surgical Procedures

The surgery pipeline at the Allen Institute is described in detail in previous reports^(40, 87). Briefly, A pre-operative injection of dexamethasone (3.2 mg kg^{-1} , subcutaneously) was administered 1 hr before surgery to reduce swelling and postoperative pain by decreasing inflammation. Mice were initially anesthetized with 5% isoflurane (1–3 min) and placed in a stereotaxic frame (Model 1900, Kopf). Isoflurane levels were maintained at 1.5–2.5% for the duration of the surgery. Body temperature was maintained at 37.5°C . Carprofen was administered for pain management ($5\text{--}10 \text{ mg/kg}$, subcutaneous), and atropine was administered to suppress bronchial secretions and regulate heart rhythm ($0.02\text{--}0.05 \text{ mg/kg}$, subcutaneous). An incision was made to remove the skin, and the exposed skull was leveled with respect to pitch (bregma–lambda level), roll, and yaw. The headframe was placed on the skull and fixed with White C&B Metabond (Parkell). Once the Metabond was dry, the mouse was placed in a custom clamp to position the skull at a rotated angle of 20° to facilitate the creation of the craniotomy over the visual cortex. A

circular piece of skull 5 mm in diameter was removed, and a durotomy was performed. The brain was covered by a 5-mm-diameter circular glass coverslip, with a 1-mm lip extending over the intact skull. The bottom of the coverslip was coated with a layer of polydimethylsiloxane (SYLGARD 184, Sigma-Aldrich) to reduce adhesion to the brain surface. The coverslip was secured to the skull with Vetbond (Patterson Veterinary). Kwik-Cast (World Precision Instruments) was added around the coverslip to further seal the implant, and Metabond bridges between the coverslip and the headframe well were created to hold the Kwik-Cast in place. At the end of the procedure, but before recovery from anesthesia, the mouse was transferred to a photodocumentation station to capture a spatially registered image of the cranial window.

Electrophysiology Experiments

The Neuropixels data was acquired at the Allen Institute as part of the OpenScope project that allows the community to apply for observation on the Allen Brain Observatory platform (<https://alleninstitute.org/division/mindscope/openscope/>). The Neuropixels pipeline at the Allen Institute is described in detail in previous reports (40, 87).

In advance of the electrophysiology experiment, intrinsic signal imaging was performed to identify visual area boundaries (88). An insertion window was designed for each mouse based on the identified visual areas. Mice were then habituated to head fixation and visual stimulation over two weeks.

On the day of recording, the cranial coverslip was removed and replaced with the insertion window containing holes aligned to six cortical visual areas. Mice were allowed to recover for 1–2 hours after the window was placed before being head-fixed in the recording rig.

Six Neuropixels probes were targeted to each of the six visual cortical areas (V1, LM, RL, AL, PM, AM). The boundaries of these areas and their retinotopic maps were obtained through intrinsic signal imaging, and insertion locations were planned to target regions responsive to the center of the LCD monitor. Probes were doused with CM-DiI (1 mM in ethanol; Thermo Fisher, V22888) for post hoc ex vivo probe localization. Each probe was mounted on a 3-axis micromanipulator (New Scale Technologies). The tip of each probe was aligned to its associated opening in the insertion window using a coordinate transformation obtained via a previous calibration procedure. XYZ manipulator coordinates were obtained via a previous calibration procedure where the tips of probes were aligned to the retinotopic centers of target visual areas using an image provided by intrinsic signal imaging. The operator then moved each probe into place with a joystick, with the probes fully retracted along the insertion axis, approximately 2.5 mm above the brain surface. The probes were manually lowered to the brain surface until spikes were visible on the electrodes closest to the tip. After the probes penetrated the brain to a depth of around 100 μm , they were inserted automatically at a rate of 200 $\mu\text{m}/\text{min}$ (total of 3.5 mm or less in the brain). After the probes reached their targets, they were allowed to settle for 5–10 min.

Neuropixels data was acquired at 30 kHz (spike band, 500 Hz high-pass filter) and 2.5 kHz (LFP band, 1,000 Hz low-pass filter) using the Open Ephys GUI (89). Videos of the eye and body were acquired at 60 Hz. Pupil size was measured via previously described means (40). Briefly, a universal eye tracking model trained in DeepLabCut (90), a ResNET-50-based network, recognizes up to 12 tracking points each around the perimeter of the eye, the pupil, and the corneal reflection. The angular velocity of the running wheel was recorded at the time of each stimulus frame, at approximately 60 Hz. All LFPs were packaged into Neurodata Without Borders (NWB) files (91) and analyzed in this paper.

Visual Stimulation Parameters

All Neuropixels recordings were according to the standardized Neuropixels visual coding pipeline. Visual stimuli were generated using custom PsychoPy scripts and displayed using an ASUS PA248Q LCD monitor with 1920x1200 pixels (55.7 cm wide, 60 Hz refresh rate). Stimuli were presented monocularly, and the monitor was positioned 15 cm from the right eye of the mouse and spanned 120 dva × 95 dva. Monitor placement was standardized across rigs such that the mouse's right eye gaze would typically be directed at the center of the monitor. Stimuli consisted of full-screen oriented drifting bar gratings, oriented 45° off the vertical (and horizontal) meridian (i.e., 45° and 135° CCW from 3 o'clock). x/X refers to one orientation and y/Y, the other orientation. Sequences consisted of 4 stimuli presented for 500 ms each with a 500 ms intervening ISI and 1 s between sequences.

Habituation to Predicted Sequence

Mice were habituated over the course of 5 behavior-only sessions in which only the predicted sequence (either xxxY or yyyX) was presented. This resulted in 2000 habituation trials before recordings.

Data Analysis

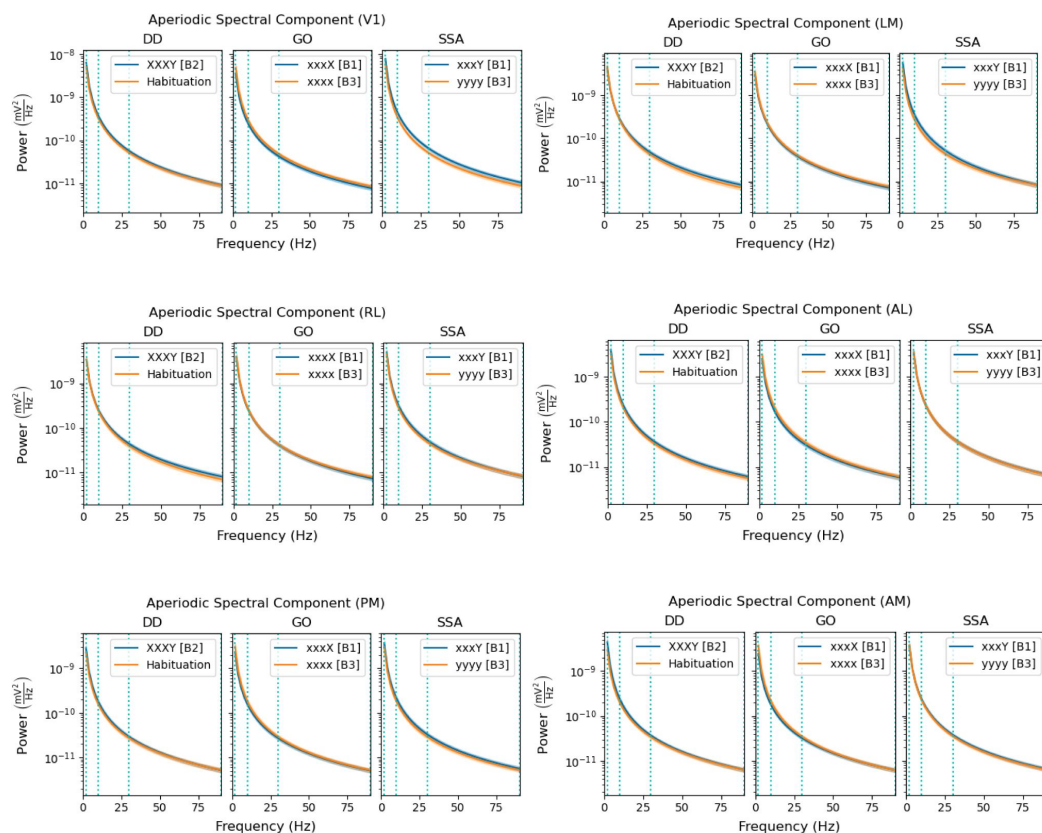
No units from 1 mouse passed the criteria due to a destabilizing event during the recording. Additionally, data from another mouse were omitted from the analysis due to issues with the visual stimulus timing. As a result, our final sample consisted of LFPs from 14 mice. Prior to running our analyses, unless otherwise stated, LFPs were first epoched according to condition within each subject's NWB file and the contiguous subset of channels histologically labelled as located in the cortex were selected. Channels were then aligned across subjects by locating the median channel within the granular layer L4 and determining the minimal distances in channels “down” into infragranular and “up” into supragranular cortex available across identically numbered probes in the same area across subjects. This channel range was used to align the LFPs for each subject to a shared channel numbering, and histological labellings were taken for the resulting cross-subject LFP data via majority vote among the per-subject probes. After alignment of the LFPs across subjects, they were appended in the trial dimension to collect within-condition, cross-subject LFP data. Finally, before calculating the spectrogram or spectrum for the LFPs, they were downsampled by 4 to obtain a single vertical line of channels.

Spectrograms were calculated by use of the [Syncopy](#) software package(92 [link](#)), used as a backend within [our own analysis code](#). The FOOOF algorithm(93 [link](#)) (version 1.1.0) was used to parameterize neural power spectra and then remove the estimated aperiodic component from spectrograms. Settings for the algorithm were set as: peak width limits : (0.5, 12.0); max number of peaks: ∞ ; minimum peak height: 0.0; peak threshold: 2.0; and aperiodic mode: “fixed”. Power spectra were parameterized across the frequency range 2 to 90 Hz with a frequency resolution of 2 Hz. Unless otherwise noted, all statistics were performed using nonparametric, cluster-based permutation testing(39 [link](#)), implemented in the [MNE](#) package(94 [link](#)), of the LFP spectrograms at the multi-channel level, with two-tailed testing and significance thresholds set to $\alpha = 0.01$.

Current source density (CSD) results came from calculating the evoked potential of oscillatory LFPs, downsampling by four (4) channels to have a single vertical column, and then employing the [Elephant](#)(95 [link](#)) package's CSD implementation. The oscillatory LFPs themselves came from replicating the analysis of van Kerkoerle et al(31 [link](#)) (among others(96 [link](#), 97 [link](#))): we first band-pass the raw (though epoched) LFPs in chosen oscillatory bands (2-10 Hz for θ , 10-30 Hz for α/β , and 30-90 Hz for γ) and obtain the oscillatory phases from [Syncopy](#)(92 [link](#)); we then selected the time and trial coordinates in the original epoched LFPs where the absolute value of the phase came

within 0.001 of π (denoting an oscillatory trough) and, after detrending of the original LFPs by subtracting its trial-averaged evoked potential, re-epoched for an oscillatory period around those timepoints in those trials. Current source density was then again calculated via Elephant.

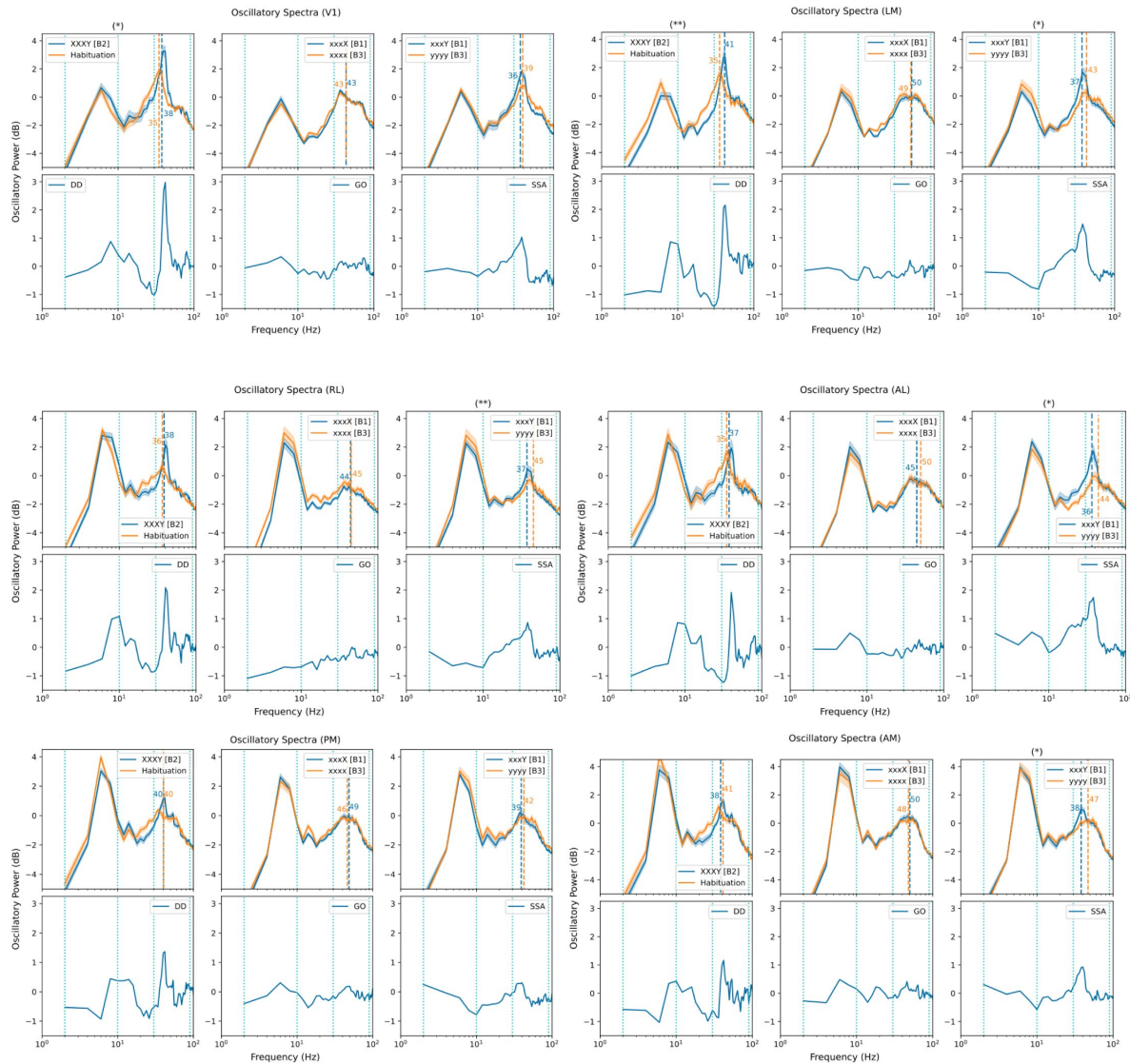
Supplementary Materials



Supplementary Figure 1

Aperiodic components of spectral responses lack significant cross-condition differences.

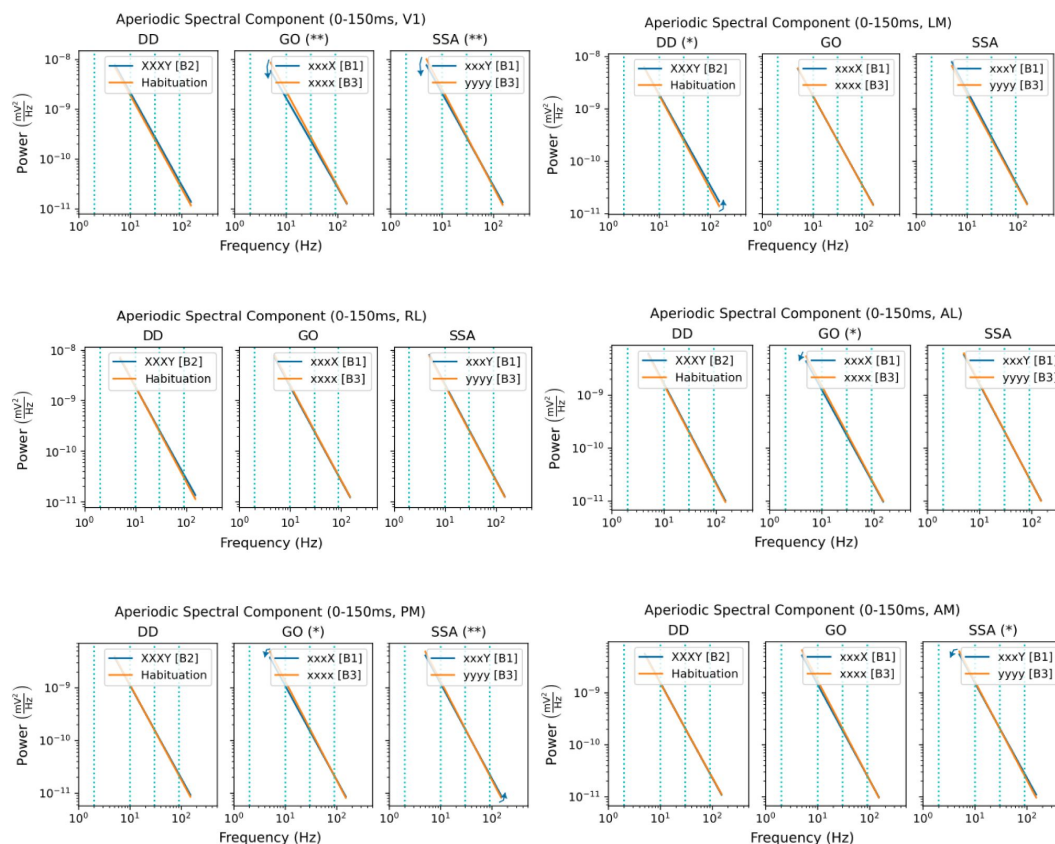
Fitting parametric models using FOOOF (93) to the trial-averaged, within-subject (N=14) spectral responses to oddballs in areas V1, LM, RL, AL, PM, and AM separated out the aperiodic component (power shown in mV^2/Hz in log-scale) from the broader spectral response; the six plots here each show this aperiodic component for the oddball (blue) and control (yellow) stimuli/sequences across the three contrasts, including error bands of two standard errors of the mean (SEMs). Across all three contrasts, the error bands for the oddball and control stimuli overlap closely enough to lack statistical significance. The blue, vertical dotted lines show the boundaries between θ (2-10 Hz), α/β (10-30 Hz), and γ (30-90 Hz) frequency bands.



Supplementary Figure 2

Oscillatory spectra show significant differences in power of α/β and γ peaks in deviance detection and stimulus-specific adaptation contrasts across areas, while global oddball contrasts show limited significant differences.

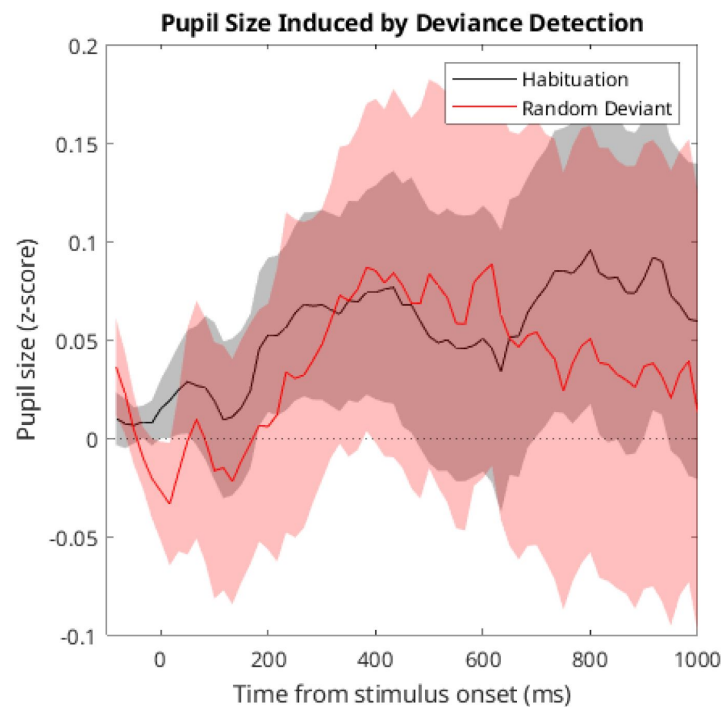
After removing the aperiodic components shown in [Supplementary Figure 1](#), the oscillatory peaks are shown here in log-space (decibels) across areas (V1, LM, RL, AL, PM, and AM) and contrasts (DD, GO, and SSA), with error-bands of two standard errors of the mean (SEMs) shown surrounding each average spectrum. DD and SSA consistently show significant (>2 SEM) differences in the heights (power above aperiodic baseline) of the γ (30-90 Hz) oscillatory peaks in V1, LM, RL, AL, and AM, with DD showing such a significant peak in PM as well. DD shows significant difference in heights of the α/β (10-30 Hz) peaks, with the control condition showing greater α/β power than the oddball condition, in V1, LM, RL, AL, PM, and AM; in LM and AL the SSA contrast shows higher α/β power in the oddball condition rather than the control condition. In V1, LM, RL, AL, and PM the GO contrast shows higher α/β power in the control condition. In DD, the oddball's γ -band peak frequency significantly (in a permutation-based t -test with 1000 permutations) shifted higher than that of the control condition V1 (35-38 Hz, $\alpha < 0.05$) and LM (35-41 Hz, $\alpha < 0.01$). In SSA, the "oddball" (having probability $P = 80\%$) had a γ -band peak frequency shifted significantly (in a permutation-based t -test with 1000 permutations) to a lower frequency in LM (37-43 Hz, $\alpha < 0.05$), RL (37-45 Hz, $\alpha < 0.01$), AL (36-44 Hz, $\alpha < 0.05$), and AM (38-47 Hz, $\alpha < 0.05$). Statistics here were calculated within conditions and across subjects, averaging over channels and trials.



Supplementary Figure 3

Aperiodic components of spectral responses in the first 150ms after oddball onset display significant cross-condition differences.

Fitting parametric models using FOOF(93) to the trial-averaged, within-subject (N=14) first 150 ms of the spectral responses to oddballs in areas V1, LM, RL, AL, PM, and AM separated out the aperiodic component (power shown in mV²/Hz in log-scale) from the broader spectral response; the six plots here each show this aperiodic component for the oddball (blue) and control (yellow) stimuli/sequences across the three contrasts, including error bands of two standard errors of the mean (SEMs). Contrasts and areas marked with an asterisk (DD-LM, GO-AL, GO-PM, SSA-AM) displayed at least one cross-condition difference in the two parameters of the aperiodic component with significance level $\alpha < 5\%$, while those marked with two asterisks (GO-V1, SSA-V1, SSA-PM) displayed at least one cross-condition difference with significance level $\alpha < 1\%$; significance was measured by a permutation-based t-test on the two independent coordinates with 1000 permutations each. The blue, vertical dotted lines show the boundaries between θ (2-10 Hz), α/β (10-30 Hz), and γ (30-90 Hz) frequency bands.



Supplementary Figure 4

Deviance detection does not significantly change pupil size.

Taking the z-scores of pupil sizes across P4 of the "XXXY" random control sequences and "xxxy" habituation sequences did not show significant differences. Solid lines show the mean z-score of pupil size across trials, while error bands show two trialwise standard deviations of the mean (trialwise standard deviation over square-root of number of trials). The dotted line shows the mean pupil size overall with a z-score of zero.

Data and Code Availability

All mouse data used in this work are openly available through the DANDI Archive (dandiset id: 000253, doi: <https://doi.org/10.48324/dandi.000253/0.240503.0152>). Software resources for the analysis of this dataset are provided here: <https://github.com/BastosLab/epych/tree/develop/notebooks/passiveglo>. The code to generate the global-local oddball sequences is provided here: <https://github.com/AllenInstitute/openscope-glo-stim>.

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Note

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97. Dougherty K., Cox M. A., Ninomiya T., Leopold D. A., Maier A. (2017) **Ongoing Alpha Activity in V1 Regulates Visually Driven Spiking Responses** *Cereb. Cortex* **27**:1113–1124 [Google Scholar](#)

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Reviewer #1 (Public review):

Summary:

The authors recorded neural activity using laminar probes while mice engaged in a global/local visual oddball paradigm. The focus of the article is on oscillatory activity, and found activity differences in theta, alpha/beta, and gamma bands related to predictability and prediction error.

I think this is an important paper, providing more direct evidence for the role of signals in different frequency bands related to predictability and surprise in the sensory cortex.

Comments:

Below are some comments that may hopefully help further improve the quality of this already very interesting manuscript.

(1) Introduction:

The authors write in their introduction: "H1 further suggests a role for θ oscillations in prediction error processing as well." Without being fleshed out further, it is unclear what role this would be, or why. Could the authors expand this statement?

(2) Limited propagation of gamma band signals:

Some recent work (e.g. [https://www.cell.com/cell-reports/fulltext/S2211-1247\(23\)00503-X](https://www.cell.com/cell-reports/fulltext/S2211-1247(23)00503-X)) suggests that gamma-band signals reflect mainly entrainment of the fast-spiking interneurons, and don't propagate from V1 to downstream areas. Could the authors connect their findings to these emerging findings, suggesting no role in gamma-band activity in communication outside of the cortical column?

(3) Paradigm:

While I agree that the paradigm tests whether a specific type of temporal prediction can be formed, it is not a type of prediction that one would easily observe in mice, or even humans. The regularity that must be learned, in order to be able to see a reflection of predictability, integrates over 4 stimuli, each shown for 500 ms with a 500 ms blank in between (and a 1000 ms interval separating the 4th stimulus from the 1st stimulus of the next sequence). In other words, the mouse must keep in working memory three stimuli, which partly occurred more than a second ago, in order to correctly predict the fourth stimulus (and signal a 1000 ms interval as evidence for starting a new sequence).

A problem with this paradigm is that positive findings are easier to interpret than negative findings. If mice do not show a modulation to the global oddball, is it because "predictive coding" is the wrong hypothesis, or simply because the authors generated a design that operates outside of the boundary conditions of the theory? I think the latter is more plausible. Even in more complex animals, (eg monkeys or humans), I suspect that participants would have trouble picking up this regularity and sequence, unless it is directly task-relevant (which it is not, in the current setting). Previous experiments often used simple pairs (where transitional probability was varied, eg, Meyer and Olson, PNAS 2012) of stimuli that were presented within an intervening blank period. Clearly, these regularities would be a lot simpler to learn than the highly complex and temporally spread-out regularity used here, facilitating the interpretation of negative findings (especially in early cortical areas, which are known to have relatively small temporal receptive fields).

I am, of course, not asking the authors to redesign their study. I would like to ask them to discuss this caveat more clearly, in the Introduction and Discussion, and situate their design in the broader literature. For example, Jeff Gavornik has used much more rapid stimulus designs and observed clear modulations of spiking activity in early visual regions. I realize that this caveat may be more relevant for the spiking paper (which does not show any spiking activity modulation in V1 by global predictability) than for the current paper, but I still think it is an important general caveat to point out.

(4) Reporting of results:

I did not see any quantification of the strength of evidence of any of the results, beyond a general statement that all reported results pass significance at an $\alpha=0.01$ threshold. It would be informative to know, for all reported results, what exactly the p-value of the significant cluster is; as well as for which performed tests there was no significant difference.

(5) Cluster test:

The authors use a three-dimensional cluster test, clustering across time, frequency, and location/channel. I am wondering how meaningful this analytical approach is. For example, there could be clusters that show an early difference at some location in low frequencies, and then a later difference in a different frequency band at another (adjacent) location. It seems a priori illogical to me to want to cluster across all these dimensions together, given that this kind of clustering does not appear neurophysiologically implausible/not meaningful. Can the authors motivate their choice of three-dimensional clustering, or better, facilitating interpretability, cluster eg at space and time within specific frequency bands (2d clustering)?

<https://doi.org/10.7554/eLife.109053.1.sa2>

Reviewer #2 (Public review):

Summary:

Sennesh and colleagues analyzed LFP data from 6 regions of rodents while they were habituated to a stimulus sequence containing a local oddball (xxxY) and later exposed to either the same (xxxY) or a deviant global oddball (xxxX). Subsequently, they were exposed to a controlled random sequence (XXXY) or a controlled deterministic sequence (xxxx or yyyy). From these, the authors looked for differences in spectral properties (both oscillatory and aperiodic) between three contrasts (only for the last stimulus of the sequence).

(1) Deviance detection: unpredictable random (XXXY) versus predictable habituation (xxxY)

(2) Global oddball: unpredictable global oddball (xxxX) versus predictable deterministic (xxxx), and

(3) "Stimulus-specific adaptation:" locally unpredictable oddball (xxxY) versus predictable deterministic (yyyy).

They found evidence for an increase in gamma (and theta in some cases) for unpredictable versus predictable stimuli, and a reduction in alpha/beta, which they consider evidence towards the "predictive routing" scheme.

While the dataset and analyses are well-suited to test evidence for predictive coding versus alternative hypotheses, I felt that the formulation was ambiguous, and the results were not very clear. My major concerns are as follows:

(1) The authors set up three competing hypotheses, in which H1 and H2 make directly opposite predictions. However, it must be noted that H2 is proposed for spatial prediction, where the predictability is computed from the part of the image outside the RF. This is different from the temporal prediction that is tested here. Evidence in favor of H2 is readily observed when large gratings are presented, for which there is substantially more gamma than in small images. Actually, there are multiple features in the spectral domain that should not be conflated, namely (i) the transient broadband response, which includes all frequencies, (ii) contribution from the evoked response (ERP), which is often in frequencies below 30 Hz, (iii) narrow-band gamma oscillations which are produced by large and continuous stimuli (which happen to be highly predictive), and (iv) sustained low-frequency rhythms in theta and alpha/beta bands which are prominent before stimulus onset and reduce after ~200 ms of stimulus onset. The authors should be careful to incorporate these in their formulation of PC, and in particular should not conflate narrow-band and broadband gamma.

(2) My understanding is that any aspect of predictive coding must be present before the onset of stimulus (expected or unexpected). So, I was surprised to see that the authors have shown the results only after stimulus onset. For all figures, the authors should show results from -500 ms to 500 ms instead of zero to 500 ms.

(3) In many cases, some change is observed in the initial ~100 ms of stimulus onset, especially for the alpha/beta and theta ranges. However, the evoked response contributes substantially in the transient period in these frequencies, and this evoked response could be different for different conditions. The authors should show the evoked responses to confirm the same, and if the claim really is that predictions are carried by genuine "oscillatory" activity, show the results after removing the ERP (as they had done for the CSD analysis).

(4) I was surprised by the statistics used in the plots. Anything that is even slightly positive or negative is turning out to be significant. Perhaps the authors could use a more stringent criterion for multiple comparisons?

(5) Since the design is blocked, there might be changes in global arousal levels. This is particularly important because the more predictive stimuli in the controlled deterministic stimuli were presented towards the end of the session, when the animal is likely less motivated. One idea to check for this is to do the analysis on the 3rd stimulus instead of the 4th? Any general effect of arousal/attention will be reflected in this stimulus.

(6) The authors should also acknowledge/discuss that typical stimulus presentation/attention modulation involves both (i) an increase in broadband power early on and (ii) a reduction in low-frequency alpha/beta power. This could be just a sensory response, without having a role in sending prediction signals per se. So the predictive routing hypothesis should involve testing for signatures of prediction while ruling out other confounds related to stimulus/cognition. It is, of course, very difficult to do so, but at the same time, simply showing a reduction in low-frequency power coupled with an increase in high-frequency power is not sufficient to prove PR.

(7) The CSD results need to be explained better - you should explain on what basis they are being called feedforward/feedback. Was LFP taken from Layer 4 LFP (as was done by van Kerkoerle et al, 2014)? The nice ">" and "<" CSD patterns (Figure 3B and 3F of their paper) in that paper are barely observed in this case, especially for the alpha/beta range.

(8) Figure 4a-c, I don't see a reduction in the broadband signal in a compared to b in the initial segment. Maybe change the clim to make this clearer?

(9) Figure 5 - please show the same for all three frequency ranges, show all bars (including the non-significant ones), and indicate the significance (p-values or by *, **, ***, etc) as done

usually for bar plots.

(10) Their claim of alpha/beta oscillations being suppressed for unpredictable conditions is not as evident. A figure akin to Figure 5 would be helpful to see if this assertion holds.

(11) To investigate the prediction and violation or confirmation of expectation, it would help to look at both the baseline and stimulus periods in the analyses.

<https://doi.org/10.7554/eLife.109053.1.sa1>

Reviewer #3 (Public review):

Summary:

In their manuscript entitled "Ubiquitous predictive processing in the spectral domain of sensory cortex", Sennesh and colleagues perform spectral analysis across multiple layers and areas in the visual system of mice. Their results are timely and interesting as they provide a complement to a study from the same lab focussed on firing rates, instead of oscillations. Together, the present study argues for a hypothesis called predictive routing, which argues that non-predictable stimuli are gated by Gamma oscillations, while alpha/beta oscillations are related to predictions.

Strengths:

(1) The study contains a clear introduction, which provides a clear contrast between a number of relevant theories in the field, including their hypotheses in relation to the present data set.

(2) The study provides a systematic analysis across multiple areas and layers of the visual cortex.

Weaknesses:

(1) It is claimed in the abstract that the present study supports predictive routing over predictive coding; however, this claim is nowhere in the manuscript directly substantiated. Not even the differences are clearly laid out, much less tested explicitly. While this might be obvious to the authors, it remains completely opaque to the reader, e.g., as it is also not part of the different hypotheses addressed. I guess this result is meant in contrast to reference 17, by some of the same authors, which argues against predictive coding, while the present work finds differences in the results, which they relate to spectral vs firing rate analysis (although without direct comparison).

(2) Most of the claims about a direction of propagation of certain frequency-related activities (made in the context of Figures 2-4) are - to the eyes of the reviewer - not supported by actual analysis but glimpsed from the pictures, sometimes, with very little evidence/very small time differences to go on. To keep these claims, proper statistical testing should be performed.

(3) Results from different areas are barely presented. While I can see that presenting them in the same format as Figures 2-4 would be quite lengthy, it might be a good idea to contrast the right columns (difference plots) across areas, rather than just the overall averages.

(4) Statistical testing is treated very generally, which can help to improve the readability of the text; however, in the present case, this is a bit extreme, with even obvious tests not reported or not even performed (in particular in Figure 5).

(5) The description of the analysis in the methods is rather short and, to my eye, was missing one of the key descriptions, i.e., how the CSD plots were baselined (which was hinted at in the

results, but, as far as I know, not clearly described in the analysis methods). Maybe the authors could section the methods more to point out where this is discussed.

(6) While I appreciate the efforts of the authors to formulate their hypotheses and test them clearly, the text is quite dense at times. Partly this is due to the compared conditions in this paradigm; however, it would help a lot to show a visualization of what is being compared in Figures 2-4, rather than just showing the results.

<https://doi.org/10.7554/eLife.109053.1.sa0>

Author response:

We would like to thank the three Reviewers for their thoughtful comments and detailed feedback. We are pleased to hear that the Reviewers found our paper to be “providing more direct evidence for the role of signals in different frequency bands related to predictability and surprise” (R1), “well-suited to test evidence for predictive coding versus alternative hypotheses” (R2), and “timely and interesting” (R3).

We perceive that the reviewers have an overall positive impression of the experiments and analyses, but find the text somewhat dense and would like to see additional statistical rigor, as well as in some cases additional analyses to be included in supplementary material. We therefore here provide a provisional letter addressing revisions we have already performed and outlining the revision we are planning point-by-point. We begin each enumerated point with the Reviewer’s quoted text and our responses to each point are made below.

Reviewer 1:

(1) Introduction:

The authors write in their introduction: “H1 further suggests a role for θ oscillations in prediction error processing as well.” Without being fleshed out further, it is unclear what role this would be, or why. Could the authors expand this statement?”

We have edited the text to indicate that theta-band activity has been related to prediction error processing as an empirical observation, and must regrettably leave drawing inferences about its functional role to future work, with experiments designed specifically to draw out theta-band activity.

(2) Limited propagation of gamma band signals:

Some recent work (e.g. [https://www.cell.com/cell-reports/fulltext/S2211-1247\(23\)00503-X](https://www.cell.com/cell-reports/fulltext/S2211-1247(23)00503-X)) suggests that gamma-band signals reflect mainly entrainment of the fast-spiking interneurons, and don't propagate from V1 to downstream areas. Could the authors connect their findings to these emerging findings, suggesting no role in gamma-band activity in communication outside of the cortical column?”

We have not specifically claimed that gamma propagates between columns/areas in our recordings, only that it synchronizes synaptic current flows between laminar layers within a column/area. We nonetheless suggest that gamma can locally synchronize a column, and potentially local columns within an area via entrainment of local recurrent spiking, to update an internal prediction/representation upon onset of a prediction error. We also point the Reviewer to our Discussion section, where we state that our results fit with a model “whereby θ oscillations synchronize distant areas, enabling them to exchange relevant signals during cognitive processing.” In our present work, we therefore remain agnostic about whether

theta or gamma or both (or alternative mechanisms) are at play in terms of how prediction error signals are transmitted between areas.

(3) Paradigm:

While I agree that the paradigm tests whether a specific type of temporal prediction can be formed, it is not a type of prediction that one would easily observe in mice, or even humans. The regularity that must be learned, in order to be able to see a reflection of predictability, integrates over 4 stimuli, each shown for 500 ms with a 500 ms blank in between (and a 1000 ms interval separating the 4th stimulus from the 1st stimulus of the next sequence). In other words, the mouse must keep in working memory three stimuli, which partly occurred more than a second ago, in order to correctly predict the fourth stimulus (and signal a 1000 ms interval as evidence for starting a new sequence).

A problem with this paradigm is that positive findings are easier to interpret than negative findings. If mice do not show a modulation to the global oddball, is it because "predictive coding" is the wrong hypothesis, or simply because the authors generated a design that operates outside of the boundary conditions of the theory? I think the latter is more plausible. Even in more complex animals, (eg monkeys or humans), I suspect that participants would have trouble picking up this regularity and sequence, unless it is directly task-relevant (which it is not, in the current setting). Previous experiments often used simple pairs (where transitional probability was varied, eg, Meyer and Olson, PNAS 2012) of stimuli that were presented within an intervening blank period. Clearly, these regularities would be a lot simpler to learn than the highly complex and temporally spread-out regularity used here, facilitating the interpretation of negative findings (especially in early cortical areas, which are known to have relatively small temporal receptive fields).

I am, of course, not asking the authors to redesign their study. I would like to ask them to discuss this caveat more clearly, in the Introduction and Discussion, and situate their design in the broader literature. For example, Jeff Gavornik has used much more rapid stimulus designs and observed clear modulations of spiking activity in early visual regions. I realize that this caveat may be more relevant for the spiking paper (which does not show any spiking activity modulation in V1 by global predictability) than for the current paper, but I still think it is an important general caveat to point out."

We appreciate the Reviewer's concern about working memory limitations in mice. Our paradigm and training followed on from previous paradigms such as Gavornik and Bear (2014), in which predictive effects were observed in mouse V1 with presentation times of 150ms and interstimulus intervals of 1500ms. In addition, we note that Jamali et al. (2024) recently utilized a similar global/local paradigm in the auditory domain with inter-sequence intervals as long as 28-30 seconds, and still observed effects of a predicted sequence (<https://elifesciences.org/articles/102702>). For the revised manuscript, we plan to expand on this in the Discussion section.

That being said, as the Reviewer also pointed out, this would be a greater concern had we not found any positive findings in our study. However, even with the rather long sequence periods we used, we did find positive evidence for predictive effects, supporting the use of our current paradigm. We agree with the reviewer that these positive effects are easier to interpret than negative effects, and plan to expand upon this in the Discussion when we resubmit.

(4) Reporting of results:

I did not see any quantification of the strength of evidence of any of the results, beyond a general statement that all reported results pass significance at an $\alpha=0.01$ threshold.

It would be informative to know, for all reported results, what exactly the p-value of the significant cluster is; as well as for which performed tests there was no significant difference."

For the revised manuscript, we can include the p-values after cluster-based testing for each significant cluster, as well as show data that passes a more stringent threshold of $p < 0.001$ (1/1000) or $p < 0.005$ (1/200) rather than our present $p < 0.01$ (1/100).

(5) Cluster test:

The authors use a three-dimensional cluster test, clustering across time, frequency, and location/channel. I am wondering how meaningful this analytical approach is. For example, there could be clusters that show an early difference at some location in low frequencies, and then a later difference in a different frequency band at another (adjacent) location. It seems a priori illogical to me to want to cluster across all these dimensions together, given that this kind of clustering does not appear neurophysiologically implausible/not meaningful. Can the authors motivate their choice of three-dimensional clustering, or better, facilitating interpretability, cluster eg at space and time within specific frequency bands (2d clustering)?"

We are happy to include a 3D plot of a time-channel-frequency cluster in the revised manuscript to clarify our statistical approach for the reviewer. We consider our current three-dimensional cluster-testing an "unsupervised" way of uncovering significant contrasts with no theory-driven assumptions about which bounded frequency bands or layers do what.

Reviewer 2:

Sennesh and colleagues analyzed LFP data from 6 regions of rodents while they were habituated to a stimulus sequence containing a local oddball (xxxy) and later exposed to either the same (xxxY) or a deviant global oddball (xxxX). Subsequently, they were exposed to a controlled random sequence (XXXY) or a controlled deterministic sequence (xxxx or yyyy). From these, the authors looked for differences in spectral properties (both oscillatory and aperiodic) between three contrasts (only for the last stimulus of the sequence).

(1) Deviance detection: unpredictable random (XXXY) versus predictable habituation (xxxy)

(2) Global oddball: unpredictable global oddball (xxxX) versus predictable deterministic (xxxx), and

(3) "Stimulus-specific adaptation:" locally unpredictable oddball (xxxy) versus predictable deterministic (yyyy).

They found evidence for an increase in gamma (and theta in some cases) for unpredictable versus predictable stimuli, and a reduction in alpha/beta, which they consider evidence towards the "predictive routing" scheme.

While the dataset and analyses are well-suited to test evidence for predictive coding versus alternative hypotheses, I felt that the formulation was ambiguous, and the results were not very clear. My major concerns are as follows:"

We appreciate the reviewer's concerns and outline how we will address them below:

(1) The authors set up three competing hypotheses, in which H1 and H2 make directly opposite predictions. However, it must be noted that H2 is proposed for spatial

prediction, where the predictability is computed from the part of the image outside the RF. This is different from the temporal prediction that is tested here. Evidence in favor of H2 is readily observed when large gratings are presented, for which there is substantially more gamma than in small images. Actually, there are multiple features in the spectral domain that should not be conflated, namely (i) the transient broadband response, which includes all frequencies, (ii) contribution from the evoked response (ERP), which is often in frequencies below 30 Hz, (iii) narrow-band gamma oscillations which are produced by large and continuous stimuli (which happen to be highly predictive), and (iv) sustained low-frequency rhythms in theta and alpha/beta bands which are prominent before stimulus onset and reduce after ~200 ms of stimulus onset. The authors should be careful to incorporate these in their formulation of PC, and in particular should not conflate narrow-band and broadband gamma."

We have clarified in the manuscript that while the gamma-as-prediction hypothesis (our H2) was originally proposed in a spatial prediction domain, further work (specifically Singer (2021)) has extended the hypothesis to cover temporal-domain predictions as well.

To address the reviewer's point about multiple features in the spectral domain: Our analysis has specifically separated aperiodic components using FOOOF analysis (Supp. Fig. 1) and explicitly fit and tested aperiodic vs. periodic components (Supp. Figs 1&2). We did not find strong effects in the aperiodic components but did in the periodic components (Supp. Fig. 2), allowing us to be more confident in our conclusions in terms of genuine narrow-band oscillations. In the revised manuscript, we will include analysis of the pre-stimulus time window to address the reviewer's point (iv) on sustained low frequency oscillations.

(2) My understanding is that any aspect of predictive coding must be present before the onset of stimulus (expected or unexpected). So, I was surprised to see that the authors have shown the results only after stimulus onset. For all figures, the authors should show results from -500 ms to 500 ms instead of zero to 500 ms.

In our revised manuscript we will include a pre-stimulus analysis and supplementary figures with time ranges from -500ms to 500ms. We have only refrained from doing so in the initial manuscript because our paradigm's short interstimulus interval makes it difficult to interpret whether activity in the ISI reflects post-stimulus dynamics or pre-stimulus prediction. Nonetheless, we can easily show that in our paradigm, alpha/beta-band activity is elevated in the interstimulus activity after the offset of the previous stimulus, assuming that we baseline to the pre-trial period.

(3) In many cases, some change is observed in the initial ~100 ms of stimulus onset, especially for the alpha/beta and theta ranges. However, the evoked response contributes substantially in the transient period in these frequencies, and this evoked response could be different for different conditions. The authors should show the evoked responses to confirm the same, and if the claim really is that predictions are carried by genuine "oscillatory" activity, show the results after removing the ERP (as they had done for the CSD analysis).

We have included an extra sentence in our Materials and Methods section clarifying that the evoked potential/ERP was removed in our existing analyses, prior to performing the spectral decomposition of the LFP signal. We also note that the FOOOF analysis we applied separates aperiodic components of the spectral signal from the strictly oscillatory ones.

In our revised manuscript we will include an analysis of the evoked responses as suggested by the reviewer.

(4) I was surprised by the statistics used in the plots. Anything that is even slightly positive or negative is turning out to be significant. Perhaps the authors could use a more stringent criterion for multiple comparisons?

As noted above to Reviewer 1 (point 4), we are happy to include supplemental figures in our resubmission showing the effects on our results of setting the statistical significance threshold with considerably greater stringency.

(5) Since the design is blocked, there might be changes in global arousal levels. This is particularly important because the more predictive stimuli in the controlled deterministic stimuli were presented towards the end of the session, when the animal is likely less motivated. One idea to check for this is to do the analysis on the 3rd stimulus instead of the 4th? Any general effect of arousal/attention will be reflected in this stimulus.

In order to check for the brain-wide effects of arousal, we plan to perform similar analyses to our existing ones on the 3rd stimulus in each block, rather than just the 4th “oddball” stimulus. Clusters that appear significantly contrasting in both the 3rd and 4th stimuli may be attributable to arousal. We will also analyze pupil size as an index of arousal to check for arousal differences between conditions in our contrasts, possibly stratifying our data before performing comparisons to equalize pupil size within contrasts. We plan to include these analyses in our resubmission.

(6) The authors should also acknowledge/discuss that typical stimulus presentation/attention modulation involves both (i) an increase in broadband power early on and (ii) a reduction in low-frequency alpha/beta power. This could be just a sensory response, without having a role in sending prediction signals per se. So the predictive routing hypothesis should involve testing for signatures of prediction while ruling out other confounds related to stimulus/cognition. It is, of course, very difficult to do so, but at the same time, simply showing a reduction in low-frequency power coupled with an increase in high-frequency power is not sufficient to prove PR.

Since many different predictive coding and predictive processing hypotheses make very different hypotheses about how predictions might be encoded in neurophysiological recordings, we have focused on prediction error encoding in this paper.

For the hypothesis space we have considered (H1-H3), each hypothesis makes clearly distinguishable predictions about the spectral response during the time period in the task when prediction errors should be present. As noted by the reviewer, a transient increase in broadband frequencies would be a signature of H3. Changes to oscillatory power in the gamma band in distinct directions (e.g., increasing or decreasing with prediction error) would support either H1 and H2, depending on the direction of change. We believe our data, especially our use of FOOOF analysis and separation of periodic from aperiodic components, coupled to the three experimental contrasts, speaks clearly in favor of the Predictive Routing model, but we do not claim we have “proved” it. This study provides just one datapoint, and we will acknowledge this in our revised Discussion in our resubmission.

(7) The CSD results need to be explained better - you should explain on what basis they are being called feedforward/feedback. Was LFP taken from Layer 4 LFP (as was done by van Kerkoerle et al, 2014)? The nice ">" and "<" CSD patterns (Figure 3B and 3F of their paper) in that paper are barely observed in this case, especially for the alpha/beta range.

We consider a feedforward pattern as flowing from L4 outwards to L2/3 and L5/6, and a feedback pattern as flowing in the opposite direction, from L1 and L6 to the middle layers.

We will clarify this in the revised manuscript.

Since gamma-band oscillations are strongest in L2/3, we re-epoched LFPs to the oscillation troughs in L2/3 in the initial manuscript. We can include in the revised manuscript equivalent plots after finding oscillation troughs in L4 instead, as well as calculating the difference in trough times within-band between layers to quantify the transmission delay and add additional rigor to our feedforward vs. feedback interpretation of the CSD data.

(8) Figure 4a-c, I don't see a reduction in the broadband signal in a compared to b in the initial segment. Maybe change the clim to make this clearer?

We are looking into the clim/colorbar and plot-generation code to figure out the visibility issue that the Reviewer has kindly pointed out to us.

*(9) Figure 5 - please show the same for all three frequency ranges, show all bars (including the non-significant ones), and indicate the significance (p-values or by *, **, ***, etc) as done usually for bar plots.*

We will add the requested bar-plots for all frequency ranges, though we note that the bars given here are the results of adding up the spectral power in the channel-time-frequency clusters that already passed significance tests and that adding secondary significance tests here may not prove informative.

(10) Their claim of alpha/beta oscillations being suppressed for unpredictable conditions is not as evident. A figure akin to Figure 5 would be helpful to see if this assertion holds.

As noted above, we will include the requested bar plot, as well as examining alpha/beta in the pre-stimulus time-series rather than after the onset of the oddball stimulus.

(11) To investigate the prediction and violation or confirmation of expectation, it would help to look at both the baseline and stimulus periods in the analyses.

We will include for the Reviewer's edification a supplementary figure showing the spectrograms for the baseline and full-trial periods to look at the difference between baseline and prestimulus expectation.

Reviewer 3:

Summary:

In their manuscript entitled "Ubiquitous predictive processing in the spectral domain of sensory cortex", Sennesh and colleagues perform spectral analysis across multiple layers and areas in the visual system of mice. Their results are timely and interesting as they provide a complement to a study from the same lab focussed on firing rates, instead of oscillations. Together, the present study argues for a hypothesis called predictive routing, which argues that non-predictable stimuli are gated by Gamma oscillations, while alpha/beta oscillations are related to predictions.

Strengths:

(1) The study contains a clear introduction, which provides a clear contrast between a number of relevant theories in the field, including their hypotheses in relation to the present data set.

(2) The study provides a systematic analysis across multiple areas and layers of the visual cortex."

We thank the Reviewer for their kind comments.

Weaknesses:

(1) It is claimed in the abstract that the present study supports predictive routing over predictive coding; however, this claim is nowhere in the manuscript directly substantiated. Not even the differences are clearly laid out, much less tested explicitly. While this might be obvious to the authors, it remains completely opaque to the reader, e.g., as it is also not part of the different hypotheses addressed. I guess this result is meant in contrast to reference 17, by some of the same authors, which argues against predictive coding, while the present work finds differences in the results, which they relate to spectral vs firing rate analysis (although without direct comparison).

We agree that in this manuscript we should restrict ourselves to the hypotheses that were directly tested. We have revised our abstract accordingly, and softened our claim to note only that our LFP results are compatible with predictive routing.

(2) Most of the claims about a direction of propagation of certain frequency-related activities (made in the context of Figures 2-4) are - to the eyes of the reviewer - not supported by actual analysis but glimpsed from the pictures, sometimes, with very little evidence/very small time differences to go on. To keep these claims, proper statistical testing should be performed.

In our revised manuscript, we will either substantiate (with quantification of CSD delays between layers) or soften the claims about feedforward/feedback direction of flow within the cortical column.

(3) Results from different areas are barely presented. While I can see that presenting them in the same format as Figures 2-4 would be quite lengthy, it might be a good idea to contrast the right columns (difference plots) across areas, rather than just the overall averages.

In our revised manuscript we will gladly include a supplementary figure showing the right-column difference plots across areas, in order to make sure to include aspects of our dataset that span up and down the cortical hierarchy.

(4) Statistical testing is treated very generally, which can help to improve the readability of the text; however, in the present case, this is a bit extreme, with even obvious tests not reported or not even performed (in particular in Figure 5).

We appreciate the Reviewer's concern for statistical rigor, and as noted to the other reviewers, we can add different levels of statistical description and describe the p-values associated with specific clusters. Regarding Figure 5, we must protest as the bar heights were computed came from clusters already subjected to statistical testing and found significant. We could add a supplementary figure which considers untested narrowband activity and tests it only in the "bar height" domain, if the Reviewer would like.

(5) The description of the analysis in the methods is rather short and, to my eye, was missing one of the key descriptions, i.e., how the CSD plots were baselined (which was hinted at in the results, but, as far as I know, not clearly described in the analysis methods). Maybe the authors could section the methods more to point out where this is discussed.

We have added some elaboration to our Materials and Methods section, especially to specify that CSD, having physical rather than arbitrary units, does not require baselining.

(6) While I appreciate the efforts of the authors to formulate their hypotheses and test them clearly, the text is quite dense at times. Partly this is due to the compared conditions in this paradigm; however, it would help a lot to show a visualization of what is being compared in Figures 2-4, rather than just showing the results.

In the revised manuscript we will add a visual aid for the three contrasts we consider.

We are happy to inform the editors that we have implemented, for the Reviewed Preprint, the direct textual Recommendations for the Authors given by Reviewers 2 and 3. We will implement the suggested Figure changes in our revised manuscript. We thank them for their feedback in strengthening our manuscript.

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